Distribution System Model Specification

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Part I

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Chapter 1

Distribution System Model & Data Specification

Welcome to the specification of a four-wire distribution-system optimal power flow (OPF) model and its accompanying data model. This site is the canonical, version-controlled home of the specification: every page is plain Markdown that evolves through pull requests rather than by editing a PDF.

It is the version-controlled successor to the LaTeX/PDF *Mathematical Model and Data Model* document, developed by the IEEE PES Task Force on Benchmarking Multiconductor OPF (BMOPF) for Distribution Systems.

Download as PDF

The complete specification is also available as a single, print-ready document: **DistributionSystemModelSpecification.pdf**.

1.1 New here? Start with the Overview

Then work through the foundations before diving into individual components.

1.2 Foundations

The shared model that every component page builds on.

- Overview — scope and structure of the specification
- Background & scope — what is (and isn't) modelled, and why
- Notation — symbols and mathematical conventions
- Data input formatting — how a case is described
- Grounding — neutral, earth, and return paths
- Worked example — a small four-bus network end to end
- Document metadata — provenance and versioning fields

1.3 Components

Per-element data models and equations.

- Buses · Lines · Switches
- Loads · Generators · Shunts · Capacitors
- Voltage sources · Transformers

1.4 Optimisation

- Objective & feasibility

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1.6 Contributing

Corrections and refinements are welcome — see the Contributing guide. The "Edit on GitHub" link on any page takes you to its source file, and every push to `main` rebuilds and publishes this site automatically. Notable changes are recorded in the Changelog.

Part II

Foundations

Chapter 2

Overview

2.1 Model specification

Status: prototype

This is a **prototype** of a self-contained model + data specification spanning the full network model: foundations (Background & scope, Notation, Data input formatting, Grounding, Worked example, Document metadata); every component (Buses, Lines, Switches, Loads, Generators, Shunts, Capacitors, Voltage sources, Transformers); and the Objective and feasibility formulation. Remaining work is refinement and Task-Force review rather than new components.

Purpose

This is the mathematical and data-model specification for four-wire distribution system optimal power flow (OPF), developed by the IEEE PES Task Force on Benchmarking Multiconductor OPF (BMOPF) for Distribution Systems. It is the version-controlled successor to the LaTeX/PDF *Mathematical Model and Data Model* document, and evolves through pull requests rather than by editing a PDF.

Two principles govern every page:

1. **The specification is the source of truth.** The equations, symbols, bounds, and data-model field names on these pages, together with the JSON Schema, form the normative contract that datasets and tools implement.
2. **Everything is in SI.** Voltages in volts, currents in amperes, impedances in ohms, admittances in siemens, powers in watts/var/VA, angles in radians. Per-unit scaling is a numerical convenience a solver may apply internally; it is **out of scope** for this specification, and no per-unit quantity appears here.

How each component page is organised

Every component page follows the same five-part structure, so a reader always knows where to find the data fields, the physics, and the bounds.

The **cartesian vs engineering** split (part 5) is deliberate. A *cartesian bound* constrains the real and imaginary components of a decision variable directly — a rectangle in the complex plane, cheap and convex, used mainly to bound the search. An *engineering bound* constrains a quantity an engineer cares about — a voltage magnitude, a thermal current, a sequence-component unbalance — and is generally a circle (quadratic) or an angle sector (bilinear). Conflating the two hides which limits are physical and which are numerical hygiene.

Part	Question it answers
1. Data model	Which JSON fields describe this component, with types, units, and required/optional status.
2. Input symbols	The mathematical symbol each field maps to (a *parameter*).
3. Variables	The unknowns this component introduces (a *variable*).
4. Equality constraints	The device physics — what the component *is*.
5. Inequality constraints	The bounds, split into **cartesian variable bounds** (box bounds on a variable's own components) and **engineering bounds** (physically meaningful magnitude/angle limits).

The foundational model is stated in **complex phasors**. A solver typically works in **rectangular real** variables — each complex equation split into its real and imaginary parts, each complex variable becoming two real variables — but that realisation is a downstream implementation choice and is not part of this specification.

Model summary

The complete feasible set at a glance — the objective, every bound, and every device constraint, with the page that defines each. Bounds are optional (an absent bound is not enforced); constraints are always active for the elements present.

Category	Item	Page
Objective	Minimise active-power dispatch cost	Objective
Voltage bounds	Phase-to-ground, -neutral, -phase, sequence, neutral cap, angle	Buses
Current bounds	Line / switch / transformer thermal, generator current	Lines, Switches, Generators
Power bounds	Generator P·Q box + apparent-power circle; transformer rating; line apparent power	Generators, Transformers
KVL / Ohm's law	Line series drop + π-shunt	Lines
KCL	Nodal current balance	Buses
Device behaviour	Load & generator power; transformer winding pairs; switch state; shunt/capacitor admittance current	Loads, Generators, Transformers, Switches, Shunts, Capacitors
Reference / grounding	Voltage-source fixing; perfect & impedance grounding	Voltage sources, Grounding

Reading order

Start with the **foundations**:

1. **Background & scope** — why the specification exists, what it covers, and the design choices behind it.
2. **Notation** — typography (variables vs parameters, real vs complex), the complex-phasor symbols and their rectangular realisation, transform matrices, the element-wise bound idiom, and the set/topology machinery used on every later page.
3. **Data input formatting** — units, how complex numbers and matrices are encoded in JSON, and required-vs-optional field semantics.

4. **Grounding** — the common-reference ground model that the component pages apply locally.
5. **Worked example** — every set constructed from one small network, to make the abstract machinery concrete.

Then the **components** (start with Buses and Lines), the Objective and feasibility formulation, and the Document metadata. The Modelling notes & FAQ collects recurring case-building questions, and References & further reading points to the textbooks and papers behind the model.

Self-containment

This section links only within itself, so the whole `spec/` folder can be moved to another repository without dangling references.

Chapter 3

Background & scope

3.1 Background and scope

Why this specification exists, what it does and does not cover, and the design choices behind it. This page frames the rest of the specification; it introduces no model content.

Motivation

Distribution networks are unbalanced: single-phase loads, untransposed lines, and single-phase laterals mean the phases cannot be collapsed to a positive-sequence equivalent without losing the physics. A faithful optimal power flow (OPF) for distribution must therefore reason at the level of **individual conductors**.

At the same time, the range of utility problems posed as network-constrained optimisation has grown — power flow, state estimation, volt-var control, DER scheduling, dynamic operating envelopes, optimal droop settings. Yet there is little standardisation: few openly-licensed unbalanced network models exist, so papers rely on ad-hoc modified cases and results are hard to compare. This specification, with its companion data library, provides a **common, openly-licensed model** so approaches can be compared directly — the distribution analogue of transmission-side benchmark libraries.

Scope: beyond classical OPF

Although "OPF" names this document, the goal is broader than minimising generation cost. The unifying requirement of the target problems is **an accurate conductor-level representation of an unbalanced network subject to a selectable set of bounds** — not any one objective. Bounded cost minimisation is a convenient, solver-comparable starting point. With fixed demand and one uniform non-negative price on every real-power injection, minimizing total injection is equivalent to minimizing real losses. It does not make the nonconvex feasible set or optimum unique. The same physics underpins maximum load delivery, conservation voltage reduction, dynamic operating envelopes, and state estimation.

This is why **bounds are optional throughout the data model**: different formulations activate different subsets of the feasible region. The specification is a reusable foundation, not infrastructure for a single problem.

Design choices

- **SI units.** All physical quantities in SI (see Data input formatting), so the data model commits to no per-unit base. Per-unit is a solver-internal convenience, out of scope here.

- **JSON + schema.** Parseable from any language; key–value structure lets extensions add nested entries without breaking readers. A JSON Schema provides basic structural checks.

- **Real numbers, not complex.** Every complex quantity is a pair of real fields, and the model solves in real variables — for cross-language compatibility (see Notation).
- **Explicit buses.** Buses are a first-class list with their own terminals and bounds — unlike OpenDSS, where buses are implicit in element connectivity.
- **String identifiers.** Buses, lines, loads, etc. carry unique string IDs, not forced sequential integers (unlike MATPOWER).
- **Wire coordinates.** Quantities are per-conductor ("wire"), not sequence/symmetrical components, keeping the format flexible and extendable to any wire count.

What is modelled

The specification covers the common branch and nodal elements, with these capabilities:

- **Topology:** meshed networks, electrically parallel branches, radial or looped.
- **Conductors:** 1- to 4-wire lines with full mutual coupling; explicit neutral and earth (no Kron reduction); perfect grounding and grounding through impedance.
- **Branch elements:** lines, switches, and galvanically-isolated transformers (single-phase, centre-tap, wye–delta, delta–wye, n-winding).
- **Nodal elements:** loads (constant-power and voltage-dependent ZIP/exponential), generators, shunts, capacitors, and voltage sources.
- **Impedance:** line series impedance and shunt admittance given directly, either as a shared linecode or as per-line matrices.

Present limitations

The model deliberately targets features universally required for distribution OPF, not every possible device. In this version:

- A **single voltage source** (one reference bus).
- **Snapshot** solves — no inter-temporal coupling (no storage state-of-charge dynamics, no OLTC time-domain control).
- Transformer **saturation** and detailed frequency-dependent effects are not modelled (a no-load magnetising shunt *is* supported).
- The default objective is linear generation/dispatch cost; quadratic cost terms are not included.

Relaxations beyond the original PDF

Several restrictions listed in the older Task Force PDF have been lifted in this implementation and are documented as first-class features here: voltage-dependent (ZIP/exponential) **loads**; **inline per-line impedance/admittance** matrices as an alternative to a linecode; the general **n-winding** transformer; a magnetising **no-load shunt** and internal **neutral grounding**; and **continuous tap** optimisation. Each carries a reconciliation note on its page.

Out of scope

The format is a practical, lightweight model for OPF research — it does not aim to replace the Common Information Model (CIM), and it does not prescribe solver software. Its OPF formulation was inspired by PowerModelsDistribution's `IVRENPowerModel`, but has been generalized well beyond it — native JSON data, the full set of element configurations above, and careful grounding.

Chapter 4

Notation

4.1 Notation

This page defines the notation used throughout the Model specification: the typography that distinguishes variables from parameters and real from complex, the voltage and current symbols and their rectangular realisation, the standard transform matrices, the element-wise idiom used to write magnitude bounds, and the set/topology machinery that links elements to buses.

All quantities are in **SI** units (V, A, Ω, S, W, var, VA, rad).

Typography

A symbol's *typeface* and *colour* encode its type. This lets an equation be read at a glance: whether a quantity is known input data or a solved unknown, and whether it is real or complex.

Symbol	Meaning
x	real scalar **variable**
$\mathbf{x}$	real vector/matrix **variable**
x	complex scalar **variable**
$\mathbf{x}$	complex vector/matrix **variable**
x	real scalar **parameter**
$\mathbf{x}$	real vector/matrix **parameter**
x	complex scalar **parameter**
$\mathbf{x}$	complex vector/matrix **parameter**
x	string **parameter**
$\mathbf{x}$	array of string **parameters**
$\mathcal{X}$	set

Operators and accessors:

Colour in this web rendering

Colour is applied to **symbol and parameter definitions** and to the headline equations, exactly as in the Task Force PDF. In long derivations colour is sometimes dropped for legibility; the type of any symbol is always its type at definition. Nothing about the model depends on colour — it is a reading aid.

Symbol	Meaning
$\mathbf{X}^{\mathsf{T}}$	transpose
$\mathbf{X}^{*}$	(element-wise) complex conjugate
$\mathbf{X}^{\mathsf{H}}$	conjugate transpose
$\circ$	element-wise (Hadamard) product
j	imaginary unit, $j^2 = -1$
$a\angle b$	polar form $a\,e^{jb}$
$\Re(\cdot),\ \Im(\cdot)$	real / imaginary part
$\mathbf{1},\ \mathbf{0}$	all-ones / all-zeros vector

Voltage: complex phasor and its rectangular realisation

The primary voltage quantity at bus i is a **complex, stacked per-terminal vector**. For a four-terminal bus with phases a, b, c and neutral n,

$$\mathbf{U}_i = \begin{bmatrix} U_{i,a} \\ U_{i,b} \\ U_{i,c} \\ U_{i,n} \end{bmatrix} \in \mathbb{C}^{|\mathcal{N}_i|},$$

where $U_{i,p}$ is the voltage of terminal p **to ground** (ground is a single 0 V reference across the network). The stacking order is the bus's declared terminal order (see Terminal names).

The complex vector decomposes into real components in rectangular or polar form:

$$\mathbf{U}_i = \mathbf{U}_i^{\Re} + j\,\mathbf{U}_i^{\Im} = \mathbf{U}_i^{\mathsf{mag}} \angle\, \boldsymbol{\theta}_i,$$

with $\mathbf{U}_i^{\Re}, \mathbf{U}_i^{\Im} \in \mathbb{R}^{|\mathcal{N}_i|}$ the real and imaginary parts (black — real variables), $\mathbf{U}_i^{\mathsf{mag}}$ the magnitude and $\boldsymbol{\theta}_i$ the angle.

The foundational model on each page is written with the **complex** vectors. The implementation solves in the **rectangular real** parts — one variable per part per terminal — so every complex equality becomes a pair of real equalities; this realisation is described in each page's *Implementation* section, not repeated in the physics.

Phase selection. $\mathbf{U}_i[\mathcal{P}]$ selects the phase (non-neutral) sub-vector $[U_{i,a}; U_{i,b}; U_{i,c}]$.

Currents

Complex current vectors follow the same stacking. The **terminal current flowing into element** at its bus is the primary quantity; for a line ℓ from bus i toward bus j it is $\mathbf{I}_{\ell ij}$, and it splits into a series and a shunt part (see Lines):

$$\mathbf{I}_{\ell ij} = \mathbf{I}_{\ell ij}^{\mathsf{s}} + \mathbf{I}_{\ell ij}^{\mathsf{sh}}.$$

The sign convention throughout is **positive current flows into the bus** at the terminal where it is summed by Kirchhoff's current law (KCL).

Standard transforms and constants

Several fixed matrices recur. With the unit rotation $\alpha = e^{j2\pi/3}$:

Phase-to-neutral (four-terminal bus → three phase-to-neutral voltages):

$$\mathbf{M}^{Y} = \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & -1 \end{bmatrix}.$$

Phase-to-phase / delta (line-to-line differences of the three phases):

$$\mathbf{M}^{\Delta} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ -1 & 0 & 1 \end{bmatrix}.$$

Symmetrical components (Fortescue), mapping three phase quantities to zero/positive/negative sequence:

$$\mathbf{F} = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & \alpha & \alpha^2 \\ 1 & \alpha^2 & \alpha \end{bmatrix}, \qquad \mathbf{U}_i^{\text{sym}} = \begin{bmatrix} U_i^0 \\ U_i^1 \\ U_i^2 \end{bmatrix}.$$

The element-wise bound idiom

Magnitude bounds are written first as a vector inequality on magnitudes, then in an equivalent **smooth (quadratic) form** using the Hadamard product $\circ$ and the conjugate, which is what the solver receives. For a generic complex vector $\mathbf{z}$ with real bound vectors $\mathbf{z}^{\min}, \mathbf{z}^{\max}$:

$$\mathbf{z}^{\min} \leq |\mathbf{z}| \leq \mathbf{z}^{\max} \quad \Longleftrightarrow \quad \mathbf{z}^{\min} \circ \mathbf{z}^{\min} \ \leq \ \mathbf{z} \circ \mathbf{z}^{*} \ \leq \ \mathbf{z}^{\max} \circ \mathbf{z}^{\max}.$$

The upper (circle) bound is convex; a non-zero lower bound is non-convex. Component k reads $(z_k^{\min})^2 \leq \Re(z_k)^2 + \Im(z_k)^2 \leq (z_k^{\max})^2$, i.e. `vr^2 + vi^2` in the code.

This is the **engineering-bound** idiom. It is distinct from a **cartesian bound**, which constrains a variable's own real/imaginary components with a box $\underline{x} \leq \Re(z_k) \leq \overline{x}$ (and likewise for $\Im$) — a rectangle, not a circle. Both appear in part 5 of each component page and are kept separate.

Sets and indices

Finite sets collect the network's elements; each element is referenced by a unique string ID.

The **element sets** each collect one kind of network element:

$$\mathcal{N}_i \subseteq \mathcal{N}$$

denotes the terminals of bus i. (Libraries — linecodes, wire data, line geometries — are referenced by string id, not collected as topological sets.)

From these, **derived sets** capture how elements attach to buses. There are four families — topology, connectivity, terminal mappings, and configurations — summarised in one table at the end of this section; the prose below defines each family, and $\mathcal{T}, \mathcal{C}, \mathcal{M}, \mathcal{R}$ are reserved for them (distinct from the element sets above).

Symbol	Represents	Index	Alt. index
$\mathcal{P}$	phases, e.g. $\{a, b, c\}$	p	q
$\mathcal{N}$	terminals (nodes), e.g. $\mathcal{P} \cup \{n\}$	p	q
$\mathcal{F}$	configurations, $\{$WYE, DELTA, SINGLE_PHASE$\}$	f	
$\mathcal{I}$	buses	i	j
$\mathcal{L}$	lines	ℓ	
$\mathcal{X}$	transformers	x	
$\mathcal{W}$	switches	w	
$\mathcal{S}$	voltage sources	s	
$\mathcal{G}$	generators	g	
$\mathcal{D}$	loads (demand)	d	
$\mathcal{H}$	shunts	h	
$\mathcal{K}$	capacitors	κ	

Topology: linking branches to buses

Branch elements (lines, transformers, switches) connect two buses. A **triple index** ℓij names line ℓ oriented from bus i to bus j; the *forward* topology set and its *reverse* are

$$\ell ij \in \mathcal{T}^{L\rightarrow} \subset \mathcal{L} \times \mathcal{I} \times \mathcal{I}, \qquad \mathcal{T}^{L\leftarrow} = \{\, \ell ji \mid \ell ij \in \mathcal{T}^{L\rightarrow} \,\}, \qquad \mathcal{T}^{L} = \mathcal{T}^{L\rightarrow} \cup \mathcal{T}^{L\leftarrow}.$$

The triple index allows parallel branches between the same bus pair. Transformer and switch topologies $\mathcal{T}^X, \mathcal{T}^W$ are defined identically, and the **network topology** is their union, $\mathcal{T} = \mathcal{T}^L \cup \mathcal{T}^X \cup \mathcal{T}^W$ (likewise for the forward and reverse senses).

Connectivity: linking nodal elements to buses

Nodal elements attach to a single bus, indexed by (element, bus):

$$di \in \mathcal{C}^D \subset \mathcal{D} \times \mathcal{I}, \quad gi \in \mathcal{C}^G, \quad hi \in \mathcal{C}^H, \quad ri \in \mathcal{C}^V, \quad \kappa i \in \mathcal{C}^K, \quad si \in \mathcal{C}^S.$$

This model version permits a single voltage source, so the source bus set $\mathcal{I}^{\text{source}} = \{i : si \in \mathcal{C}^S\}$ has $|\mathcal{I}^{\text{source}}| = 1$.

Terminal names and maps

A bus declares an **ordered** list of terminal names $\mathbf{N}_i$ (`terminal_names`), e.g. $[$`"a"`, `"b"`, `"c"`, `"n"`$]$. Element vectors and matrices stack in this order. Terminal names are strings; common conventions include $\{a, b, c, n\}$, $\{1, 2, 3, n\}$, and IEC $\{L1, L2, L3, N\}$.

Every element carries a **terminal map** — the string array $\mathbf{N}_{\ell i}$ (`terminal_map_from`), $\mathbf{N}_{\ell j}$ (`terminal_map_to`) for a line — listing which of its bus's terminals each conductor connects to, so per-phase properties align across the network. As a *set*, a terminal mapping is written $\mathcal{M}^{\bullet}$: a member $d\,z\,p$ of the load mapping $\mathcal{M}^D$ says load d's conductor at order z maps to bus terminal p; branch mappings carry an extra bus index (ℓizp).

The **neutral terminal** n of a bus is identified by the bus's declaration (an explicit neutral field, or a terminal named `"n"`/`"N"`). If a bus has no neutral, all its terminals are treated as phases.

Configurations

Loads and generators declare a connection **configuration** $f \in \mathcal{F}$ (WYE / DELTA / SINGLE_PHASE). The configuration sets pair each element with its configuration: $df \in \mathcal{R}^D$ for loads, $gf \in \mathcal{R}^G$ for generators.

Grounding

Ground is a single 0 V reference. Terminals listed in a bus's `perfectly_grounded_terminals` form the ground mapping $ip \in \mathcal{M}^\emptyset \subset \mathcal{I} \times \mathcal{N}$; their voltage is fixed to zero. Lines and shunts always carry an implicit ground connection (their shunt admittance is defined to ground). See Grounding for the full model.

Overview of derived sets

All topology, connectivity, terminal-mapping, and configuration sets, with the tuple that indexes each:

Symbol	Represents	Member tuple
$\mathcal{T}^{L\rightarrow},\ \mathcal{T}^{L\leftarrow},\ \mathcal{T}^{L}$	line topology — forward, reverse, combined	ℓij
$\mathcal{T}^{X\rightarrow},\ \mathcal{T}^{X\leftarrow},\ \mathcal{T}^{X}$	transformer topology	xij
$\mathcal{T}^{W\rightarrow},\ \mathcal{T}^{W\leftarrow},\ \mathcal{T}^{W}$	switch topology	wij
$\mathcal{T}^{\rightarrow},\ \mathcal{T}^{\leftarrow},\ \mathcal{T}$	network topology (union of the above)	$\cdot ij$
$\mathcal{C}^S$	voltage-source–bus connectivity	si
$\mathcal{C}^G$	generator–bus connectivity	gi
$\mathcal{C}^D$	load–bus connectivity	di
$\mathcal{C}^H$	shunt–bus connectivity	hi
$\mathcal{C}^K$	capacitor–bus connectivity	κi
$\mathcal{I}^{\text{source}}$	source bus set ($\lvert\cdot\rvert = 1$)	i
$\mathcal{M}^S, \mathcal{M}^G, \mathcal{M}^D, \mathcal{M}^H, \mathcal{M}^V, \mathcal{M}^K$	nodal terminal mappings	$\cdot\, z\, p$
$\mathcal{M}^L, \mathcal{M}^T, \mathcal{M}^W$	branch terminal mappings	$\cdot\, i\, z\, p$
$\mathcal{M}^\emptyset$	ground terminal mapping	ip
$\mathcal{R}^D,\ \mathcal{R}^G$	load / generator configurations	$\cdot\, f$

Matrices (e.g. impedance) are stored **row-first** with an underscore delimiter: matrix entry A_{kj} is the field `A_k_j`, 1-indexed. So `R_series_1_2` is the $(1, 2)$ entry of the series-resistance matrix.

A minimal example (bus + line)

Two four-terminal buses $\mathcal{I} = \{\mathsf{A}, \mathsf{B}\}$ joined by one line $\mathcal{L} = \{\ell\}$ from `A` to `B`, each terminal mapped straight through:

$$\mathcal{T}^{L\rightarrow} = \{\ell\,\mathsf{A}\,\mathsf{B}\}, \quad \mathbf{N}_{\mathsf{A}} = \mathbf{N}_{\mathsf{B}} = [a, b, c, n], \quad \mathbf{N}_{\ell\mathsf{A}} = \mathbf{N}_{\ell\mathsf{B}} = [a, b, c, n].$$

If terminal n of bus B is perfectly grounded, $\mathcal{M}^\emptyset = \{\mathsf{B}\, n\}$ and $U_{\mathsf{B},n} = 0$. This is the running example used on the Buses and Lines pages.

Chapter 5

Data input formatting

5.1 Data input formatting

This page defines how physical quantities are represented in the JSON data model: units, the encoding of complex numbers and matrices, worked unit-conversion examples, and the meaning of required vs optional fields. It underpins every component page. Symbols are defined in Notation.

Units

All physical quantities are in **SI**, with two deliberate exceptions noted below. The data model carries no unit fields — units are fixed by the field, per this table.

Quantity	Unit	Symbol
Voltage	volt	V
Current	ampere	A
Length	metre	m
Active power	watt	W
Reactive power	volt-ampere reactive	var
Apparent power	volt-ampere	VA
Conductance, susceptance	siemens	S
Resistance, reactance	ohm	Ω
Angle	radian	rad
Cost rate	US dollar per kilowatt-hour	$/kWh

Non-SI exception. One quantity uses a customary unit for industry familiarity: **cost rate** in $/kWh (there is no SI unit of currency; 1 kWh $= 3.6 \times 10^6$ J).

Per-unit normalisation is a solver-internal convenience and is **out of scope** here — no per-unit quantity appears in the data model.

Complex numbers, vectors, and matrices

JSON has only ordered lists of real numbers, so:

- A **complex** quantity is stored as a **pair of real fields** (rectangular or polar). For example a voltage source is given by `v_magnitude` and `v_angle`; an impedance by `R_series_*` and `X_series_*`. The model uses real variables throughout for the same reason (see Notation).

- A **matrix** is stored **row-first** with an underscore-delimited, **1-indexed** key: entry A_{kj} is the field `A_k_j`. So `R_series_1_2` is the $(1,2)$ entry of the series-resistance matrix, and `G_from_2_2` the $(2,2)$ from-side shunt conductance.
- A **vector** (e.g. `v_min`, `i_max`) is a JSON array, ordered to match the element's terminal map or phase order as stated on each component page.

Conversion examples

Convert conventional power-systems quantities to SI before writing them. To allow exact cross-checks against tools that use other units (e.g. degrees), give constants to **at least 10 significant figures**, ideally full floating-point precision.

Quantity	Conventional	SI	Example → JSON
Active power	kilowatts (kW)	watts (W)	3 kW → `3000.0` (or `3.0e3`)
Angle	degrees	radians	120° → `2.0943951023931953`
Reactance	per-unit (on a Z-base)	ohms (Ω)	5 % on a 100 Ω base → `5.0` (Ω) on the winding field

Required and optional fields

Each component has required fields (listed with ✔ on its page) and optional ones. The interpretation of an **absent optional** field depends on its kind:

- **Absent constraint field** ⇒ that constraint is **unbounded**. A missing voltage upper bound means no upper bound is enforced; a missing `i_max` means no thermal limit.
- **Absent parameter field** ⇒ a **null / zero** value. A missing transformer `r_series_from` means that winding resistance is $0\ \Omega$.

This is why bounds are optional throughout the model: different problem formulations (cost OPF, maximum load delivery, CVR, state estimation) activate different subsets of the feasible region, so the data model lets each bound be present or absent independently.

Permissible strings and string arrays

Several fields are string-valued. Some are free identifiers; others are restricted to an enumeration or must reference terminals declared elsewhere.

Field	Restriction
`terminal_names` (a bus's conductor names)	Any strings; see the naming conventions below
`configuration` (load / generator / capacitor)	A nodal configuration string: `WYE`, `DELTA`, or `SINGLE_PHASE` (per element support — generators are `WYE` only)
`model` (load)	`CONSTANT_POWER`, `CONSTANT_CURRENT`, `CONSTANT_IMPEDANCE`, `ZIP`, `EXPONENTIAL`
`perfectly_grounded_terminals`	Array; every entry must be one of the bus's `terminal_names`
`terminal_map`, `terminal_map_from`, `terminal_map_to`	Array; every entry must be one of the target bus's `terminal_names`
`source` (linecode)	Free text (provenance note)

Terminal conventions

Terminal identifiers are **strings**; a component's terminal map carries no ordering semantics beyond the array order. The canonical written form is `"1"`, `"2"`, `"3"`, `"n"`, but the following conventions are recognised on read (case-insensitively):

Convention	Phases	Neutral
OpenDSS numeric	`"1" "2" "3"`	`"4"` (positional)
Canonical	`"1" "2" "3"`	`"n"`
Letter	`"a" "b" "c"`	`"n"` / `"N"`
IEC 60445	`"L1" "L2" "L3"`	`"N"`

Because a label alone does not say which conductor is a phase and which is the neutral, a case may classify roles once for the whole network in an optional top-level `terminal_conventions` block:

```
"terminal_conventions": { "phase": ["a", "b", "c"], "neutral": ["n"] }
```

When present the block is **authoritative** — labels are matched exactly, including case — and is always emitted on export. The role lists must be disjoint; the ground reference stays implicit. When the block is **absent**, roles are inferred: a terminal `n`/`N` is the neutral; terminal `"4"` is the neutral when the bus terminal set is exactly `{1, 2, 3, 4}`; every other terminal is a phase.

Chapter 6

Grounding

6.1 Grounding

Grounding is the subtle heart of a four-wire model: the neutral and earth are explicit, and how each element connects to ground determines the return paths for current. This page consolidates the grounding model that the component pages apply locally. Symbols are defined in Notation.

Ground as a common reference

Ground is a single 0 V reference — a "copper plate" shared across the entire network. It is not a per-element node: elements connect to a **bus terminal**, and that terminal may (or may not) connect to ground. Because ground is a common reference, no separate ground node is indexed in element matrices, and the terminal name **"g" is reserved** for the common ground across all elements.

Two ways a terminal reaches ground

- **Perfect grounding** — a bus terminal listed in `perfectly_grounded_terminals` is pinned to the reference, $U_{i,p} = 0$. Current can still flow into earth there: a free earth-injection current balances the terminal's KCL (see Buses). This is a **bus property**.
- **Grounding through an impedance** — modelled as a shunt (or capacitor) between the terminal and ground, e.g. a single-entry admittance $Y_{h,nn}$ on the neutral grounds it through $1/Y_{h,nn}$.

An **ungrounded** terminal has neither; its voltage floats, determined by the network equations. A bus whose terminals are all ungrounded needs no grounding data.

Which elements connect to ground

The "always" elements (lines, shunts, voltage sources) carry an implicit ground connection and need no grounding declaration. The only element that *optionally* declares grounding is the **bus** — everything else routes through a bus terminal or is intrinsic to the element.

In the mathematical model

Perfect grounding contributes the equality $U_{i,p} = 0$ for every $ip \in \mathcal{M}^{\emptyset}$ (the ground map) and a free earth current so KCL still balances. Impedance grounding contributes a linear admittance current $Y\,U$ to KCL, exactly like any shunt. No grounding scheme reduces the conductor matrices — neutrals and earth conductors stay explicit, which is what makes this a genuinely four-wire formulation.

Element	Connection to ground	Note
Load	never	Grounding is a bus property or a shunt
Generator	never	Grounding is a bus property or a shunt
Capacitor	never	Grounding is a bus property or a shunt
Transformer	never	Winding neutral grounding via `r/x_neutral_*` is *internal*; external grounding is a bus/shunt
Switch	never	Switches are not grounded
Bus terminal	**optional**	`perfectly_grounded_terminals` for perfect grounding; a shunt for impedance grounding
Line	**always (implicit)**	A non-zero shunt admittance passes current to ground
Shunt	**always (implicit)**	The admittance is defined between terminals and ground
Voltage source	**always**	Source voltages are defined line-to-ground

Chapter 7

Worked example

7.1 A worked network example

The Notation page defines the sets, topology, connectivity, and terminal-map machinery abstractly. This page makes them concrete by constructing every set from one small network — the fastest way to internalise how a case is assembled.

The network has four buses $\{\mathtt{A}, \mathtt{B}, \mathtt{C}, \mathtt{D}\}$ joined by a line, a transformer, and a switch, feeding single- and three-phase loads, a single-phase generator, a neutral-grounding shunt, and a voltage source, with a mix of grounded and ungrounded terminals. The circled numbers on the figure are element IDs.

Element sets

The branch and nodal elements (see the classification into branch vs nodal):

$$\begin{aligned}
&\mathcal{I} = \{\mathtt{A}, \mathtt{B}, \mathtt{C}, \mathtt{D}\}, \quad &&\mathcal{L} = \{2\}, \quad &&\mathcal{W} = \{9\}, \quad &&\mathcal{X} = \{6\}, \\
&\mathcal{D} = \{1, 5, 7\}, \quad &&\mathcal{G} = \{3\}, \quad &&\mathcal{H} = \{4\}, \quad &&\mathcal{S} = \{8\}.
\end{aligned}$$

Topology (branches → buses)

The forward topology orients each branch from its `from` bus to its `to` bus:

$$\mathcal{T}^{L\rightarrow} = \{2\,\mathtt{A}\,\mathtt{B}\}, \quad \mathcal{T}^{X\rightarrow} = \{6\,\mathtt{C}\,\mathtt{B}\}, \quad \mathcal{T}^{W\rightarrow} = \{9\,\mathtt{C}\,\mathtt{D}\}.$$

The combined line topology adds the reverse orientation, $\mathcal{T}^{L} = \{2\,\mathtt{A}\,\mathtt{B},\ 2\,\mathtt{B}\,\mathtt{A}\}$, and likewise for the transformer and switch.

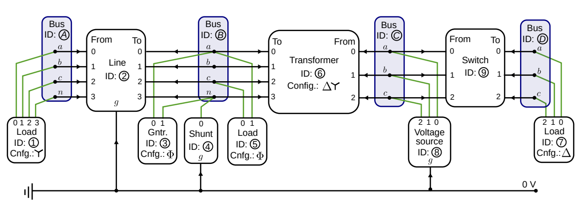

Figure 7.1: Example four-bus network with single- and three-phase elements.

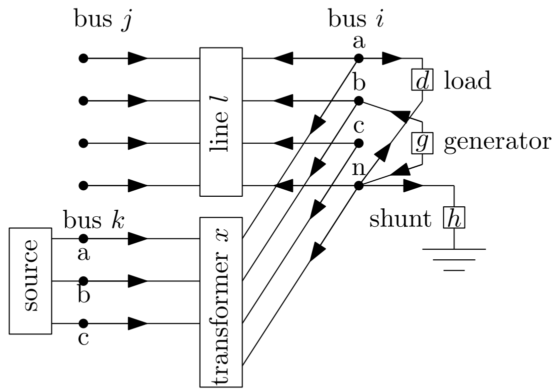

Figure 7.2: Terminal maps align an element's conductors to bus terminals.

Connectivity (nodal elements → buses)

$$\mathcal{C}^D = \{1\,\texttt{A},\ 5\,\texttt{B},\ 7\,\texttt{D}\},\quad \mathcal{C}^G = \{3\,\texttt{B}\},\quad \mathcal{C}^H = \{4\,\texttt{B}\},\quad \mathcal{C}^S = \{8\,\texttt{C}\}.$$

The single voltage source fixes $|\mathcal{I}^{\text{source}}| = 1$ (bus C is the reference).

Terminal maps

Each element lists which bus terminals its conductors connect to, in order. As stored in the data model (`terminal_map`, or `terminal_map_from/_to` for branches):

Element	Map	Notes
Load 1 (bus A)	`["a","b","c","n"]`	three-phase wye
Load 5 (bus B)	`["c","a"]`	single-phase, across c–a
Load 7 (bus D)	`["a","b","c"]`	delta
Generator 3 (bus B)	`["a","n"]`	single-phase
Shunt 4 (bus B)	`["n"]`	neutral grounding
Source 8 (bus C)	`["a","b","c"]`	
Line 2	from `["a","b","c","n"]`, to `["a","b","c","n"]`	four-wire
Transformer 6	from `["a","b","c","n"]` (C, wye), to `["a","b","c"]` (B, delta)	ΔY
Switch 9	from `["a","b","c"]`, to `["a","b","c"]`	

Configurations

The nodal-element connection configurations:

$$\mathcal{R}^D = \{1\!:\text{WYE},\ 5\!:\text{SINGLE\_PHASE},\ 7\!:\text{DELTA}\},\qquad \mathcal{R}^G = \{3\!:\text{SINGLE\_PHASE}\}.$$

Grounding

Buses declare which terminals are perfectly grounded. In this network the source bus neutral and any explicitly-earthed terminals are pinned to 0 V (bus property), while the shunt at bus B grounds the neutral through an impedance. See Grounding for the full model; the line and shunt additionally carry the implicit ground connection of their shunt admittances.

From these sets, the component pages instantiate the variables and constraints: a voltage vector per bus terminal, series currents on the line and transformer windings, load/generator/source currents, and one KCL equation per terminal tying them together.

Chapter 8

Document metadata

8.1 Document metadata

Every case may carry a top-level `meta` object providing provenance, licensing, and versioning. It is optional for backward compatibility, but conformant writers should populate it. This is a *data-model* page — `meta` carries no OPF variables or constraints. Symbols are defined in Notation.

The `meta` object

Field	Type	Description
`$schema`	string (URI)	URI of the JSON Schema this file was validated against
`version`	string	Version of this **dataset** (not the schema)
`title`	string	Human-readable dataset name (distinct from the machine-readable root `name`)
`description`	string	Free-text notes or caveats
`created`	string (ISO 8601)	Creation timestamp, UTC
`modified`	string (ISO 8601)	Last-revision timestamp, UTC
`license`	string	SPDX identifier (e.g. `CC-BY-4.0`) or an `https://` URI to the licence
`frequency`	number (Hz)	Nominal system frequency — **check-only** (validated against `line_geometry`/linecode derivation frequency; never rescales)
`authors`	object[]	Ordered authors — see below
`sources`	object[]	Upstream data references — see below
`generator`	object	Tool that wrote the file — see below
`provenance`	object	Free-form conversion/audit notes written by tooling

`authors[]`: `name`, `email`, `orcid` (bare hyphenated form `XXXX-XXXX-XXXX-XXXX`, without the `https://orcid.org/` prefix).

`sources[]`: `name`, `url` (dereferenceable `https://`), `format` (e.g. `OpenDSS`, `PMD-JSON`, `MATPOWER`), `doi`, `version`.

`generator`: `tool`, `version`.

Conformance notes

- `$schema` follows the JSON Schema `$schema` keyword convention; it should be a versioned, dereferenceable URI (a floating reference to `main` is acceptable in draft).

- Dates must be UTC and should include a time component (`THH:MM:SSZ`); a date-only value is permitted when the time is unknown.
- `license` should be a URI when the licence needs specific attribution language beyond an SPDX identifier.
- Unknown fields may be present; conformant readers must ignore them.
- `frequency` is a consistency check only — it never defaults or rescales any quantity.

Example

```
{
  "name": "enwl_network1_feeder1",
  "meta": {
    "$schema": "https://example.org/bmopf/schema/v1/bmopf.json",
    "version": "1.0",
    "title": "ENWL LV Network 1, Feeder 1",
    "description": "Four-wire 230/400 V LV feeder from the ENWL LVNS dataset.",
    "created": "2026-06-15T10:00:00Z",
    "license": "https://creativecommons.org/licenses/by/4.0/",
    "frequency": 50.0,
    "authors": [
      { "name": "A. Researcher", "email": "a@example.org", "orcid": "0000-0001-9534-2265" }
    ],
    "sources": [
      { "name": "ENWL LVNS dataset", "url": "https://www.enwl.co.uk/lvns", "format": "OpenDSS" }
    ],
    "generator": { "tool": "example-writer", "version": "0.1.0" }
  },
  "bus": { "...": {} },
  "line": { "...": {} }
}
```

Part III

Components

Chapter 9

Buses

9.1 Buses

A **bus** is a set of electrical terminals sharing a location. It owns the network's voltage variables and is where Kirchhoff's current law is enforced. Parts 1–5 state the foundational (physics) model. Symbols are defined in Notation.

1. Data model

A bus is an entry of the top-level bus object, keyed by its string ID i.

Field	Type	Unit	Req.	Description
`terminal_names`	string[]	–	✔	Ordered terminal names $\mathbf{N}_i$
`perfectly_grounded_terminals`	string[]	–		Terminals fixed to 0 V
`v_min`, `v_max`	number[]	V		Phase-to-ground magnitude bounds, one per phase terminal
`vn_max`	number	V		Neutral-to-ground magnitude cap
`vpn_min`, `vpn_max`	number[]	V		Phase-to-neutral magnitude bounds, one per phase
`vpp_min`, `vpp_max`	number[]	V		Phase-to-phase magnitude bounds, one per phase pair
`vpos_min`, `vpos_max`	number	V		Positive-sequence magnitude bounds (three-phase buses)
`vneg_max`	number	V		Negative-sequence magnitude cap (lower bound is always 0)
`vzero_max`	number	V		Zero-sequence magnitude cap (lower bound is always 0)

All bound fields are optional: an absent bound means the corresponding limit is not enforced.

2. Input symbols

3. Variables

Each terminal $p \in \mathcal{N}_i$ has a complex voltage-to-ground $U_{i,p}$, stacked into the bus voltage vector

$$\mathbf{U}_i = \left[\, U_{i,p} \,\right]_{p \in \mathcal{N}_i} \in \mathbb{C}^{|\mathcal{N}_i|}.$$

Field	Symbol	Notes
`terminal_names`	$\mathbf{N}_i$	stacking order for all bus vectors
`v_min`, `v_max`	$\mathbf{U}_i^{\min}$, $\mathbf{U}_i^{\max}$	per phase, to ground
`vn_max`	$U_{i,n}^{\max}$	scalar, neutral to ground
`vpn_min`, `vpn_max`	$\mathbf{U}_i^{Y,\min}$, $\mathbf{U}_i^{Y,\max}$	per phase, to neutral
`vpp_min`, `vpp_max`	$\mathbf{U}_i^{\Delta,\min}$, $\mathbf{U}_i^{\Delta,\max}$	per phase pair
`vpos_min`, `vpos_max`	$U_i^{1,\min}$, $U_i^{1,\max}$	positive-sequence magnitude
`vneg_max`, `vzero_max`	$U_i^{2,\max}$, $U_i^{0,\max}$	negative-/zero-sequence caps (lower bound 0)

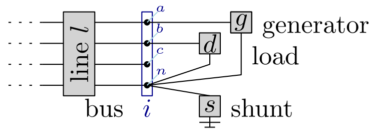

Figure 9.1: Kirchhoff's current law at a bus terminal: the signed currents of all incident elements sum to zero.

Ground is the common 0 V reference, so these are the only quantities needed to describe the bus's electrical state.

4. Equality constraints

Perfect grounding

A perfectly grounded terminal is pinned to the ground reference:

$$U_{i,p} = 0 \qquad \forall\, ip \in \mathcal{M}^{\emptyset}.$$

Voltage source (reference bus)

The voltage source s at bus i fixes its phase terminals to the reference magnitude/angle and its neutral to ground:

$$U_{i,p} = U_{i,p}^{s} = |U_{i,p}^{s}| \angle \theta_{i,p}^{s}, \qquad U_{i,n} = 0.$$

The source also injects a free slack current into KCL, making this the power-flow reference bus (detailed on the future *Voltage sources* page).

Kirchhoff's current law

At each terminal, the currents of all incident elements sum to zero (sign convention: into the bus positive):

$$\underbrace{\sum_{\ell ij \in \mathcal{T}^L} \mathbf{I}_{\ell ij}}_{\text{lines}} + \underbrace{\sum_{xij \in \mathcal{T}^X} \mathbf{I}_{xij}}_{\text{transformers}} + \underbrace{\sum_{wij \in \mathcal{T}^W} \mathbf{I}_{wij}}_{\text{switches}} + \underbrace{\sum_{di \in \mathcal{C}^D} \mathbf{I}_{d}}_{\text{loads}} - \underbrace{\sum_{gi \in \mathcal{C}^G} \mathbf{I}_{g}}_{\text{generators}} + \underbrace{\sum_{hi \in \mathcal{C}^H} \mathbf{I}_{h}}_{\text{shunts}} = \mathbf{0}.$$

This holds at every terminal except where the bus is voltage-source-fixed or grounded (there the terminal voltage is set directly, and the balancing current is a free variable rather than a constraint).

5. Inequality constraints

Cartesian variable bounds

None. The physics places no box on the rectangular components of $\mathbf{U}_i$; voltage is constrained only by the engineering bounds below and by grounding/source fixing. A box on the real/imaginary parts would impose an axis-aligned magnitude-and-angle limit with no operational meaning.

Engineering bounds

Applied at ungrounded, non-source phase terminals (the neutral is excluded from phase bounds — its voltage is set by physics, not operational limits).

Phase-to-ground magnitude. With $U_{i,n}^{\min} = 0$ and the neutral's upper bound supplied by `vn_max`:

$$\mathbf{U}_i^{\min} \circ \mathbf{U}_i^{\min} \ \le\ \mathbf{U}_i \circ \mathbf{U}_i{}^* \ \le\ \mathbf{U}_i^{\max} \circ \mathbf{U}_i^{\max}.$$

Neutral-to-ground cap (`vn_max`, only when the neutral floats):

$$|U_{i,n}| \le U_{i,n}^{\max} \iff U_{i,n}\, U_{i,n}{}^* \le (U_{i,n}^{\max})^2.$$

Phase-to-neutral magnitude. With $\mathbf{U}_i^Y = \mathbf{M}^Y\, \mathbf{U}_i$ (or $\mathbf{U}_i[\mathcal{P}]$ when the neutral is grounded):

$$\mathbf{U}_i^{Y,\min} \circ \mathbf{U}_i^{Y,\min} \ \le\ \mathbf{U}_i^Y \circ (\mathbf{U}_i^Y)^* \ \le\ \mathbf{U}_i^{Y,\max} \circ \mathbf{U}_i^{Y,\max}.$$

Phase-to-phase magnitude. With $\mathbf{U}_i^\Delta = \mathbf{M}^\Delta\, \mathbf{U}_i[\mathcal{P}]$:

$$\mathbf{U}_i^{\Delta,\min} \circ \mathbf{U}_i^{\Delta,\min} \ \le\ \mathbf{U}_i^\Delta \circ (\mathbf{U}_i^\Delta)^* \ \le\ \mathbf{U}_i^{\Delta,\max} \circ \mathbf{U}_i^{\Delta,\max}.$$

Symmetrical-component magnitudes (three-phase buses). The sequence voltages use phase-to-neutral inputs when a neutral floats, phase-to-ground otherwise:

$$\mathbf{U}_i^{\text{sym}} = \mathbf{F}\, \mathbf{U}_i^Y = \begin{bmatrix} U_i^0 \\ U_i^1 \\ U_i^2 \end{bmatrix}, \qquad \begin{gathered} (U_i^{1,\min})^2 \le U_i^1 (U_i^1)^* \le (U_i^{1,\max})^2, \\ U_i^2 (U_i^2)^* \le (U_i^{2,\max})^2, \qquad U_i^0 (U_i^0)^* \le (U_i^{0,\max})^2. \end{gathered}$$

Only the positive sequence carries a lower bound.

Intra-bus angle difference. For each phase pair (p, q), with a nominal offset $\Delta = \theta_{i,q}^{\text{nom}} - \theta_{i,p}^{\text{nom}}$ and $z = U_{i,p}{}^* U_{i,q}\, e^{-j\Delta} = c + j s$:

$$\tan(\theta_i^{\Delta,\min})\, c \ \le\ s \ \le\ \tan(\theta_i^{\Delta,\max})\, c,$$

which bounds the angle between the two terminals (faithful while $c > 0$, i.e. the centred deviation stays within $\pm\pi/2$).

Chapter 10

Lines

10.1 Lines

A **line** is a multi-conductor power cable or overhead line, modelled as a nominal Π equivalent: a series impedance with a shunt admittance half-section at each end. Parts 1–5 state the foundational (physics) model. Symbols are defined in Notation.

1. Data model

A line is an entry of the top-level `line` object, keyed by its string ID ℓ. It carries **exactly one impedance source**: either a referenced `linecode` (per-metre matrices scaled by `length`) *or* inline absolute matrices.

The `oneOf` schema rule requires **either** `linecode` + `length` **or** at least `R_series_1_1` + `X_series_1_1` inline (and then *no* `linecode`).

Linecode

A `linecode` c stores **per-metre** matrices shared across lines of the same type.

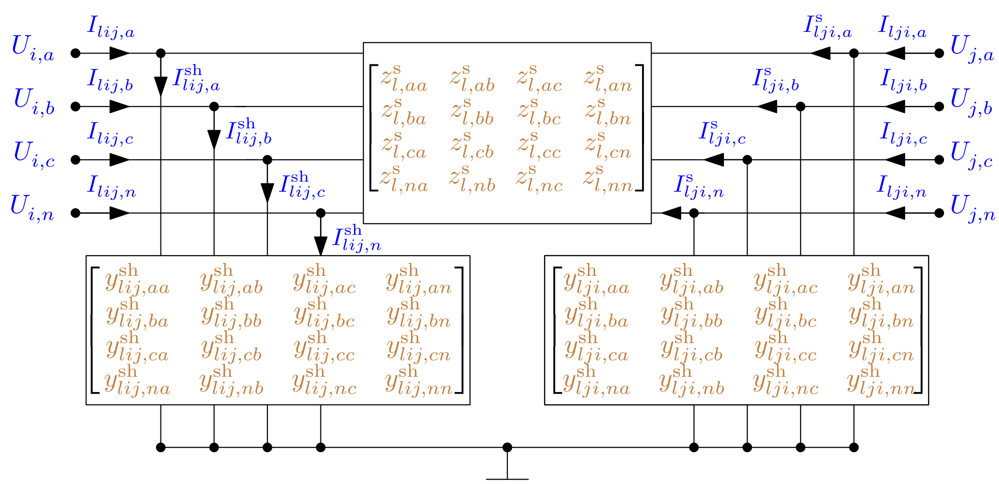

Figure 10.1: Four-wire nominal Π-model of a line: a series impedance with a shunt admittance half-section at each end, both referenced to ground.

Field	Type	Unit	Req.	Description
`bus_from`, `bus_to`	string	–	✔	Endpoint bus IDs i, j
`terminal_map_from`	string[]	–	✔	Conductor→terminal map at `bus_from`, $\mathbf{N}_{\ell i}$
`terminal_map_to`	string[]	–	✔	Conductor→terminal map at `bus_to`, $\mathbf{N}_{\ell j}$
`linecode`	string	–	(one-of)	Linecode ID (per-metre matrices)
`length`	number	m	(with `linecode`)	Line length L_ℓ
`R_series_k_j`, `X_series_k_j`	number	Ω	(one-of)	Inline **absolute** series impedance entries
`G_from_k_j`, `B_from_k_j`	number	S		Inline from-side shunt admittance entries
`G_to_k_j`, `B_to_k_j`	number	S		Inline to-side shunt admittance entries
`i_max`	number[]	A		Per-conductor current-magnitude limit (overrides linecode)
`s_max`	number[]	VA		Per-conductor apparent-power limit (overrides linecode)

Field	Type	Unit	Req.	Description
`R_series_k_j`, `X_series_k_j`	number	Ω/m	✔	Per-metre series impedance $\Re, \Im(\mathbf{Z}^{\mathrm{s}}_c)$
`G_from_k_j`, `G_to_k_j`	number	S/m		Per-metre shunt conductance
`B_from_k_j`, `B_to_k_j`	number	S/m		Per-metre shunt susceptance
`i_max`	number[]	A		Per-conductor current limit $\mathbf{I}^{\max}_c$
`s_max`	number[]	VA		Per-conductor apparent-power limit
`source`	string	–		Free-text provenance note for the impedance/admittance matrices (e.g. `fem`, `datasheet`, `import`); descriptive metadata only, no effect on the electrical model

2. Input symbols

Field	Symbol	Notes
linecode `R_series`, `X_series`	$\mathbf{Z}^{\mathrm{s}}_c = \mathbf{R} + j\mathbf{X}$	Ω/m matrix
linecode `G/B_from`, `G/B_to`	$\mathbf{Y}^{\mathrm{sh}}_c = \mathbf{G} + j\mathbf{B}$	S/m matrix
`length`	L_ℓ	metres
line `R_series`/`X_series`, `G/B_*`	$\mathbf{Z}^{\mathrm{s}}_\ell$, $\mathbf{Y}^{\mathrm{sh}}_{\ell ij}$, $\mathbf{Y}^{\mathrm{sh}}_{\ell ji}$	Ω, S (absolute)
`i_max`	$\mathbf{I}^{\max}_{\ell ij}$	per conductor
`s_max`	$\mathbf{S}^{\max}_{\ell ij}$	per conductor

Parameter construction

The line's series impedance and shunt admittances come from **one** source:

$$\text{linecode:} \quad \mathbf{Z}^{\mathsf{s}}_{\ell} = \mathbf{Z}^{\mathsf{s}}_{c}\, L_{\ell}, \qquad \mathbf{Y}^{\mathsf{sh}}_{\ell ij} = \mathbf{Y}^{\mathsf{sh}}_{\ell ji} = \tfrac{1}{2}\, \mathbf{Y}^{\mathsf{sh}}_{c}\, L_{\ell};$$

or the inline absolute matrices are used directly (never scaled by length). Inline data permits **independent** from- and to-side shunts $\mathbf{Y}^{\mathsf{sh}}_{\ell ij} \neq \mathbf{Y}^{\mathsf{sh}}_{\ell ji}$.

3. Variables

A line has $n_\ell = |\mathbf{N}_{\ell i}|$ conductors. The **series current** flowing from i toward j is the independent unknown:

$$\mathbf{I}^{\mathsf{s}}_{\ell ij} \in \mathbb{C}^{n_\ell}.$$

The **terminal current** $\mathbf{I}_{\ell ij}$ (what enters the bus's KCL) and the **shunt current** $\mathbf{I}^{\mathsf{sh}}_{\ell ij}$ are the other line currents; the equalities of part 4 relate them to $\mathbf{I}^{\mathsf{s}}_{\ell ij}$ and the bus voltages.

4. Equality constraints

Ohm's law (series voltage drop)

Across the series impedance, for the forward orientation ℓij:

$$\mathbf{U}_j[\mathbf{N}_{\ell j}] = \mathbf{U}_i[\mathbf{N}_{\ell i}] - \mathbf{Z}^{\mathsf{s}}_{\ell}\, \mathbf{I}^{\mathsf{s}}_{\ell ij}.$$

The impedance matrix is full: off-diagonal entries couple the voltage drop on one conductor to the current in another.

Shunt currents

Each Π half-section draws a current set by its admittance and the local bus voltage:

$$\mathbf{I}^{\mathsf{sh}}_{\ell ij} = \mathbf{Y}^{\mathsf{sh}}_{\ell ij}\, \mathbf{U}_i, \qquad \mathbf{I}^{\mathsf{sh}}_{\ell ji} = \mathbf{Y}^{\mathsf{sh}}_{\ell ji}\, \mathbf{U}_j.$$

Terminal current and series-current conservation

The current entering the line at each bus is the series current plus that end's shunt current, and the two directional series currents are equal and opposite:

$$\mathbf{I}_{\ell ij} = \mathbf{I}^{\mathsf{s}}_{\ell ij} + \mathbf{I}^{\mathsf{sh}}_{\ell ij}, \qquad \mathbf{I}_{\ell ji} = \mathbf{I}^{\mathsf{s}}_{\ell ji} + \mathbf{I}^{\mathsf{sh}}_{\ell ji}, \qquad \mathbf{I}^{\mathsf{s}}_{\ell ij} + \mathbf{I}^{\mathsf{s}}_{\ell ji} = \mathbf{0}.$$

The terminal currents $\mathbf{I}_{\ell ij}, \mathbf{I}_{\ell ji}$ are the contributions this line makes to KCL at buses i and j (see Buses §4).

5. Inequality constraints

Cartesian variable bounds

A **box** may be placed on the series-current variable $\mathbf{I}^{\mathsf{s}}_{\ell ij}$ to bound the search. Because the engineering limit constrains the *total* (terminal) current, the box on the series part is the limit inflated by the worst-case shunt contribution ρ_k on that row:

$$|\Re(I^{\mathsf{s}}_{\ell ij,k})| \leq I^{\max}_{\ell ij,k} + \rho_k, \qquad |\Im(I^{\mathsf{s}}_{\ell ij,k})| \leq I^{\max}_{\ell ij,k} + \rho_k,$$

with $\rho_k \leq \sum_{j'} |Y^{\mathsf{sh}}_{\ell ij,kj'}|\, U^{\max}_{i,j'}$ from the endpoint voltage caps. This is a solver-conditioning aid; it is implied by the engineering bound below.

Engineering bounds

Thermal current limit on the **terminal current**, at both ends, for all conductors including neutral:

$$\mathbf{I}_{\ell ij} \circ (\mathbf{I}_{\ell ij})^* \leq \mathbf{I}^{\max}_{\ell ij} \circ \mathbf{I}^{\max}_{\ell ij}, \qquad \mathbf{I}_{\ell ji} \circ (\mathbf{I}_{\ell ji})^* \leq \mathbf{I}^{\max}_{\ell ij} \circ \mathbf{I}^{\max}_{\ell ij}.$$

Apparent-power limit (optional, from s_max), with the terminal power $\mathbf{S}_{\ell ij} = \mathbf{U}_i \circ (\mathbf{I}_{\ell ij})^*$:

$$\mathbf{S}_{\ell ij} \circ (\mathbf{S}_{\ell ij})^* \leq \mathbf{S}^{\max}_{\ell ij} \circ \mathbf{S}^{\max}_{\ell ij}.$$

Per-line angle difference (optional `va_diff_min`/`va_diff_max`). For each conductor, with $z = U_{i,\cdot}\,(U_{j,\cdot})^* = c + js$:

$$\tan(\theta^{\Delta,\min}_{\ell})\, c \;\leq\; s \;\leq\; \tan(\theta^{\Delta,\max}_{\ell})\, c.$$

Chapter 11

Switches

11.1 Switches

A **switch** is an ideal, lossless branch connecting two buses conductor-for-conductor. When *closed* it short-circuits its terminals; when *open* it carries no current. Symbols are defined in Notation.

1. Data model

A switch is an entry of the top-level `switch` object, keyed by its string ID w.

Field	Type	Unit	Req.	Description
`bus_from`, `bus_to`	string	–	✔	Endpoint bus IDs i, j
`terminal_map_from`	string[]	–	✔	Conductor→terminal map at `bus_from`
`terminal_map_to`	string[]	–	✔	Conductor→terminal map at `bus_to`
`open_switch`	bool	–	✔	`true` = open (no current), `false` = closed
`i_max`	number[]	A		Per-conductor current-magnitude limit

2. Input symbols

Field	Symbol	Notes
`open_switch`	state $\in \{\text{open}, \text{closed}\}$	selects which equality applies
`i_max`	$\mathbf{I}_{wij}^{\max}$	per conductor

3. Variables

A switch has n_w conductors and one complex current per conductor, flowing from i toward j:

$$\mathbf{I}_{wij} \in \mathbb{C}^{n_w}.$$

The reverse current $\mathbf{I}_{wji}$ is its negative.

4. Equality constraints

Closed switch — zero voltage drop

A closed switch equates the two ends conductor-by-conductor:

$$\mathbf{U}_i[\mathbf{N}_{wi}] = \mathbf{U}_j[\mathbf{N}_{wj}].$$

Open switch — zero current

An open switch carries no current and imposes no voltage coupling (the two buses are electrically disconnected at these conductors):

$$\mathbf{I}_{wij} = \mathbf{0}.$$

Current conservation and KCL

In both states the current is conserved, and enters each bus's KCL with opposite sign:

$$\mathbf{I}_{wji} = -\mathbf{I}_{wij}.$$

5. Inequality constraints

Cartesian variable bounds

When a current limit is present, a **box** is placed on the switch-current variable components, $|\Re(I_{wij,k})|,\ |\Im(I_{wij,k})| \leq I^{\max}_{wij,k}$, to bound the search. It is implied by the engineering limit below.

Engineering bounds

Thermal current limit (a switch has no shunt, so both ends carry equal magnitude — one constraint suffices):

$$\mathbf{I}_{wij} \circ (\mathbf{I}_{wij})^* \ \leq\ \mathbf{I}^{\max}_{wij} \circ \mathbf{I}^{\max}_{wij}.$$

Chapter 12

Loads

12.1 Loads

A **load** draws a specified power at a bus. Its power may be constant, or vary with voltage (ZIP / exponential). Three connection configurations are supported: single-phase, wye (with neutral return), and delta. Symbols are defined in Notation.

1. Data model

A load is an entry of the top-level `load` object, keyed by its string ID d.

Field	Type	Unit	Req.	Description
`bus`	string	–	✔	Host bus ID i
`terminal_map`	string[]	–	✔	Conductor→terminal map $\mathbf{N}_d$
`configuration`	string	–	✔	`WYE`, `SINGLE_PHASE`, or `DELTA`
`p_nom`, `q_nom`	number[]	W, var	✔	Nominal per-sub-load active/reactive power
`model`	string	–		`CONSTANT_POWER` (default), `CONSTANT_CURRENT`, `CONSTANT_IMPEDANCE`, `ZIP`, `EXPONENTIAL`
`v_nom`	number[]	V	(zip/exp)	Nominal voltage at which `p_nom`/`q_nom` hold
`alpha_z`, `alpha_i`, `alpha_p`	number[]	–	(zip)	Active ZIP fractions ($\alpha_Z + \alpha_I + \alpha_P = 1$)
`beta_z`, `beta_i`, `beta_p`	number[]	–	(zip)	Reactive ZIP fractions
`gamma_p`, `gamma_q`	number[]	–	(exp)	Active/reactive voltage exponents

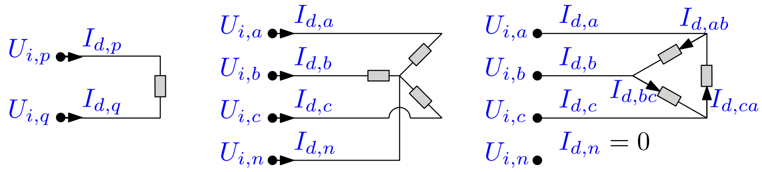

Figure 12.1: Single-phase (left), wye (centre), and delta (right) load connections and their current variables.

2. Input symbols

Field	Symbol	Notes
p_nom, q_nom	P_d^{nom}, Q_d^{nom}	per sub-load
v_nom	V_d^{nom}	reference voltage
ZIP fractions	$\alpha_Z, \alpha_I, \alpha_P$, $\beta_Z, \beta_I, \beta_P$	
exponents	γ_P, γ_Q	

A **sub-load** k is one two-terminal power element of the load: a phase-to-neutral branch (WYE), the single terminal pair (SINGLE_PHASE), or one line-to-line branch (DELTA).

3. Variables

Each sub-load k draws a complex current $I_{d,k}$, stacked into $\mathbf{I}_d$. These are the currents the load contributes to KCL; the neutral return current is not an independent variable.

4. Equality constraints

Sub-load voltage drop and power

Let $\Delta U_{d,k}$ be the voltage across sub-load k: phase-to-neutral for WYE, the terminal-pair difference for SINGLE_PHASE, line-to-line for DELTA. The complex power the sub-load absorbs is

$$S_{d,k} = \Delta U_{d,k}\,(I_{d,k})^* = P_{d,k} + j\,Q_{d,k}.$$

Constant power (default) pins it to the nominal setpoint:

$$P_{d,k} = P_{d,k}^{\text{nom}}, \qquad Q_{d,k} = Q_{d,k}^{\text{nom}}.$$

Voltage-dependent (ZIP) scales the setpoint by impedance/current/power fractions of the voltage-magnitude ratio $|\Delta U_{d,k}|/V_{d,k}^{\text{nom}}$:

$$P_{d,k} = P_{d,k}^{\text{nom}}\left(\alpha_Z \frac{|\Delta U_{d,k}|^2}{(V_{d,k}^{\text{nom}})^2} + \alpha_I \frac{|\Delta U_{d,k}|}{V_{d,k}^{\text{nom}}} + \alpha_P\right),$$
$$Q_{d,k} = Q_{d,k}^{\text{nom}}\left(\beta_Z \frac{|\Delta U_{d,k}|^2}{(V_{d,k}^{\text{nom}})^2} + \beta_I \frac{|\Delta U_{d,k}|}{V_{d,k}^{\text{nom}}} + \beta_P\right).$$

Exponential uses a power law $P_{d,k} = P_{d,k}^{\text{nom}}\,(|\Delta U_{d,k}|/V_{d,k}^{\text{nom}})^{\gamma_P}$ (and likewise Q with γ_Q). Constant-impedance, constant-current and constant-power are the special cases $\alpha = (1,0,0), (0,1,0), (0,0,1)$ and integer exponents $\gamma \in \{2,1,0\}$.

Current conservation

The sub-load currents sum to zero over the load's terminals (the neutral / return carries $-\sum_k I_{d,k}$), giving the KCL contribution at each host terminal.

5. Inequality constraints

None (foundational). A load has no rating bounds in this model — it is a fixed or voltage-dependent power sink.

Chapter 13

Generators

13.1 Generators

A **generator** injects a *dispatchable* power at a bus — active and reactive power lie within bounds rather than being fixed (a fixed injection is modelled as a negative load). It shares the load's bilinear power form. Symbols are defined in Notation.

1. Data model

A generator is an entry of the top-level `generator` object, keyed by its string ID g.

Field	Type	Unit	Req.	Description
`bus`	string	–	✔	Host bus ID i
`terminal_map`	string[]	–	✔	Conductor→terminal map $\mathbf{N}_g$
`configuration`	string	–	✔	`WYE` (only supported configuration)
`p_min`, `p_max`	number[]	W		Per-phase active-power bounds
`q_min`, `q_max`	number[]	var		Per-phase reactive-power bounds
`s_max`	number[]	VA		Per-phase apparent-power rating
`i_max`	number[]	A		Per-conductor current-magnitude limit (incl. optional neutral entry)
`cost`	number[]	$/kWh		Per-phase linear generation cost

Only the `WYE` configuration is supported for generators. Delta and single-phase generation should be modelled another way — e.g. as separate wye or single-phase injections via loads (negative loads).

2. Input symbols

3. Variables

Each phase conductor k injects a complex current $I_{g,k}$ (the current sent *to* the bus, opposite sign to a load), stacked into $\mathbf{I}_g$. The neutral return is implicit in KCL.

Field	Symbol	Notes
`p_min`, `p_max`	$P_g^{\min}$, $P_g^{\max}$	per phase
`q_min`, `q_max`	$Q_g^{\min}$, $Q_g^{\max}$	per phase
`s_max`	$\mathbf{S}_g^{\max}$	per phase
`i_max`	$\mathbf{I}_g^{\max}$	per conductor (last entry may bound the neutral return)
`cost`	$\mathbf{c}_g$	per phase

4. Equality constraints

With $\Delta U_{g,k}$ the sub-generator voltage (phase-to-neutral for the `WYE` configuration), the injected complex power is

$$S_{g,k} = \Delta U_{g,k}\, (I_{g,k})^{*} = P_{g,k} + j\, Q_{g,k}.$$

Current conservation over the generator's terminals gives its KCL contribution (injection positive at the phase terminal, return at the neutral).

5. Inequality constraints

Cartesian variable bounds

When `i_max` is present, a **box** on the current components $|\Re(I_{g,k})|,\ |\Im(I_{g,k})| \leq I_{g,k}^{\max}$ bounds the search; implied by the current circle below.

Engineering bounds

Active/reactive power box (the dispatch range):

$$P_{g,k}^{\min} \leq P_{g,k} \leq P_{g,k}^{\max}, \qquad Q_{g,k}^{\min} \leq Q_{g,k} \leq Q_{g,k}^{\max}.$$

Apparent-power circle (optional):

$$P_{g,k}^{2} + Q_{g,k}^{2} \leq (S_{g,k}^{\max})^{2}.$$

Current-magnitude circle (optional), per conductor:

$$I_{g,k}(I_{g,k})^{*} \leq (I_{g,k}^{\max})^{2}.$$

For a star-connected generator whose `i_max` carries a trailing neutral entry, the neutral return current $-\mathbf{1}^{\mathsf{T}}\mathbf{I}_g[\mathcal{P}]$ is additionally bounded by that entry.

Chapter 14

Shunts

14.1 Shunts

A **shunt** is a fixed admittance connected between a set of bus terminals (and, implicitly, ground). It models capacitor banks, grounding impedances, and similar passive elements. Symbols are defined in Notation.

1. Data model

A shunt is an entry of the top-level `shunt` object, keyed by its string ID h.

Field	Type	Unit	Req.	Description
`bus`	string	–	✔	Host bus ID i
`terminal_map`	string[]	–	✔	Conductor→terminal map $\mathbf{N}_h$
`G_k_j`	number	S	✔ (G_1_1)	Conductance matrix entries
`B_k_j`	number	S	✔ (B_1_1)	Susceptance matrix entries

2. Input symbols

Field	Symbol	Notes
`G_k_j`, `B_k_j`	$\mathbf{Y}_h = \mathbf{G} + j\mathbf{B}$	admittance matrix (S), row-first entries

The matrix is stored row-first: entry (k, j) is field `G_k_j` / `B_k_j`. Ground is never indexed (its voltage is zero); zero-filled rows/columns should be removed by trimming the terminal map.

3. Variables

None. A shunt introduces no unknown — its current is determined entirely by the bus voltage and its fixed admittance.

4. Equality constraints

The current drawn into the shunt at its terminals is linear in the bus voltage:

$$\mathbf{I}_h = \mathbf{Y}_h \, \mathbf{U}_i[\mathbf{N}_h].$$

This current is the shunt's contribution to KCL at bus i (leaving the bus toward the admittance/ground). Examples: a single-entry $Y_{h,nn}$ on terminal n grounds the neutral through an impedance; a delta pattern $Y_{\text{cap}}\,\mathbf{M}^{\Delta}$ on $\{a, b, c\}$ models a delta capacitor bank.

5. Inequality constraints

None. With a fixed admittance the current follows the voltage; a current limit would risk trivial infeasibility and is not imposed.

Chapter 15

Capacitors

15.1 Capacitors

A **capacitor** is a fixed, nameplate-rated shunt capacitor bank. Electrically it is a constant susceptance derived from its rating, delivering voltage-dependent reactive power. Symbols are defined in Notation.

1. Data model

A capacitor is an entry of the top-level `capacitor` object, keyed by its string ID.

Field	Type	Unit	Req.	Description
bus	string	–	✔	Host bus ID i
terminal_map	string[]	–	✔	Conductor→terminal map
configuration	string	–	✔	WYE, SINGLE_PHASE, or DELTA
q_nom	number[]	var	✔	Rated reactive power per phase (WYE) or per phase-pair (DELTA)
v_nom	number	V	✔	Rated voltage (phase-to-neutral for WYE, line-to-line for DELTA)

2. Input symbols

Field	Symbol	Notes
q_nom	$\mathbf{Q}^{\text{rated}}$	per phase / phase-pair
v_nom	V^{nom}	rated voltage

The bank's susceptance follows from the nameplate: $B = Q^{\text{rated}}/(V^{\text{nom}})^2$, assembled (per configuration) into a terminal-space admittance $\mathbf{Y} = j\,\mathbf{B}$.

3. Variables

None. Like a shunt, a capacitor introduces no unknown — its current follows from the bus voltage and its fixed susceptance.

4. Equality constraints

The current drawn into the capacitor is linear in the bus voltage, and is the bank's contribution to KCL at bus i:

$$\mathbf{I} = j\,\mathbf{B}\,\mathbf{U}_i[\mathbf{N}].$$

The delivered reactive power is therefore voltage-dependent, $\mathbf{B}\,|\mathbf{U}|^2$ — it rises and falls with the square of the terminal voltage, the defining behaviour of a fixed capacitor (as opposed to a fixed-power source).

5. Inequality constraints

None. With a fixed susceptance the current follows the voltage; no rating bound is imposed.

Chapter 16

Voltage sources

16.1 Voltage sources

A **voltage source** is an ideal voltage reference with a current slack: it fixes the bus terminal voltages and injects whatever current the rest of the network requires. It is the power-flow reference (slack) bus. This model version permits exactly one. Symbols are defined in Notation.

1. Data model

A voltage source is an entry of the top-level `voltage_source` object, keyed by its string ID s.

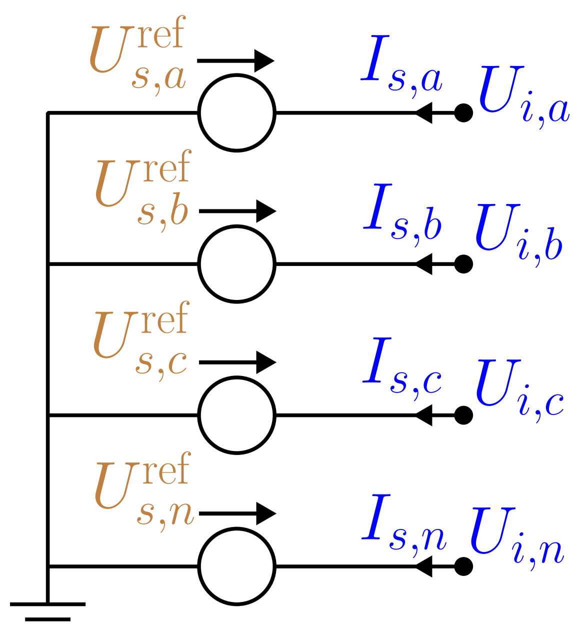

Figure 16.1: Voltage source: a fixed line-to-ground voltage reference with a free slack current.

Field	Type	Unit	Req.	Description
bus	string	–	✔	Host bus ID i
terminal_map	string[]	–	✔	Conductor→terminal map $\mathbf{N}_s$ (phase terminals)
v_magnitude	number[]	V	✔	Per-terminal voltage magnitude
v_angle	number[]	rad	✔	Per-terminal voltage angle

Field	Symbol	Notes
v_magnitude	$\lvert\mathbf{U}_s^s\rvert$	per terminal
v_angle	$\boldsymbol{\theta}_s^s$	per terminal

2. Input symbols

3. Variables

The source injects a **slack current** $I_{s,k}$ per phase terminal, stacked into $\mathbf{I}_s$. It is otherwise unconstrained — it absorbs whatever power balance the network requires, which is what makes this the reference bus.

4. Equality constraints

Fixed reference voltage

Each phase terminal is fixed to its polar reference, and the source-bus neutral to ground:

$$U_{i,p} = |U^s_{s,p}| \angle \theta^s_{s,p}, \qquad U_{i,n} = 0.$$

Slack current injection

The slack current enters KCL at the phase terminals, with its return at the neutral: $+I_{s,k}$ at phase t_k, and $-\sum_k I_{s,k}$ at the neutral.

5. Inequality constraints

Cartesian variable bounds

None — the slack current is free.

Engineering bounds

None (foundational). A pure slack has no rating.

Chapter 17

Transformers

17.1 Transformers

A **transformer** couples two buses through **galvanically isolated** windings — primary and secondary share no conductor, only magnetic flux. Because the winding topologies differ qualitatively, each configuration is a distinct data-model object: `single_phase`, `center_tap`, `wye_delta`, `delta_wye`, and the general `n_winding`. Every model is built from an idealised winding pair plus three loss components. Symbols are defined in Notation.

What a "winding" counts — ports, not physical coils

In the literature and in tools, a **winding** (and an *n-winding* transformer) usually counts the number of **buses / voltage levels the transformer connects** — its *ports* — not the number of physical coils. For a *single-phase* transformer the two coincide: a two-winding single-phase unit has two coils and two ports. For a *three-phase* transformer they do **not**: a three-phase two-winding transformer has **six** physical coils (three per side) yet is still called "two-winding". Throughout this specification "winding" means a **port** — one bus connection at one voltage level — and each winding comprises n_phase physical coils. The `n_winding` object's `windings` array lists these ports (each with its own bus), so n is the number of buses, and per phase there are n coils on the shared core.

1. Data model

Each subtype is an entry under `transformer.<subtype>`, keyed by its string ID x. Common fields (two-winding subtypes):

Terminal-map lengths: `single_phase` 2 + 2; `center_tap` 2 (from) + 3 (to); `wye_delta` 4 (wye) + 3 (delta); `delta_wye` 3 (delta) + 4 (wye). The `n_winding` object instead carries a `windings` array (each with `bus`, `terminal_map`, `v_ref`, `configuration`, `r_winding`, optional `delta_roll`, `i_max`) and pairwise short-circuit reactances `x_sc` keyed `"i_j"`.

2. Input symbols

3. Variables

Each winding conductor k on side $\sigma \in \{\text{fr}, \text{to}\}$ carries a complex winding current $I_{x,\sigma,k}$. For the winding spanning terminal pair (p_k, q_k) on bus b^σ, write the **winding voltage**

$$V^{\sigma}_{x,k} = U_{b^\sigma,p_k} - U_{b^\sigma,q_k}$$

Field	Type	Unit	Req.	Description
`bus_from`, `bus_to`	string	–	✔	Endpoint bus IDs i, j
`terminal_map_from`, `terminal_map_to`	string[]	–	✔	Conductor→terminal maps (lengths per subtype)
`v_ref_from`, `v_ref_to`	number	V	✔	Reference winding voltages — set the turns ratio
`s_rating`	number	VA	✔	Nameplate apparent-power rating
`r_series_from`, `x_series_from`	number	Ω		From-winding series leakage
`r_series_to`, `x_series_to`	number	Ω		To-winding series leakage
`g_no_load`, `b_no_load`	number	S		No-load (core-loss / magnetising) shunt
`r_neutral_from/_to`, `x_neutral_from/_to`	number	Ω		Winding-neutral grounding impedance (OpenDSS `rneut`/`xneut`)
`tap`, `tap_min`, `tap_max`	number	-		From-side tap multiplier; a free OPF variable when `tap_min < tap_max`
`i_max_from`, `i_max_to`	number[]	A		Per-conductor current limits

Field	Symbol	Notes
`v_ref_from`, `v_ref_to`	U_i^{ref}, U_j^{ref}	turns ratio $N = U_i^{\text{ref}}/U_j^{\text{ref}}$
`r/x_series_from`	$Z_x^{\text{fr}} = R_x^{\text{fr}} + jX_x^{\text{fr}}$	from-winding leakage
`r/x_series_to`	$Z_x^{\text{to}} = R_x^{\text{to}} + jX_x^{\text{to}}$	to-winding leakage
`g/b_no_load`	$Y_0 = G_0 + jB_0$	magnetising shunt
`r/x_neutral_*`	$y_n = 1/(R_n + jX_n)$	neutral grounding
`tap`	τ (fixed) or τ (variable)	$N_{\text{eff}} = N\,\tau$
`s_rating`	$S_x^{\max}$	nameplate

(phase-to-neutral for a wye winding, line-to-line for a delta winding, with $U = 0$ when q_k is absent/ground). When the tap is free, the ratio N_{eff} becomes a decision variable.

4. Equality constraints

The idealised winding pair

Every transformer is built from **ideal winding pairs** obeying flux linkage, complex- power conservation, and winding KCL. With winding EMFs E_x^{fr}, E_x^{to} and the reference voltages standing in for the turns ratio:

$$\frac{E_x^{\text{fr}}}{U_i^{\text{ref}}} = \frac{E_x^{\text{to}}}{U_j^{\text{ref}}}, \qquad U_i^{\text{ref}}\, I_{x,\text{fr}} + U_j^{\text{ref}}\, I_{x,\text{to}} = 0 \iff N_{\text{eff}}\, I_{x,\text{fr}} + I_{x,\text{to}} = 0.$$

The second relation is the **ampere-turn balance**; with all losses removed the EMF is the terminal voltage and $V_x^{\text{fr}} = N_{\text{eff}}\, V_x^{\text{to}}$.

Loss components

Three loss elements dress the ideal pair. Every subtype places them the **same** way, consistent with the OpenDSS reference model; the four subtype diagrams below are the same family of picture, differing only in

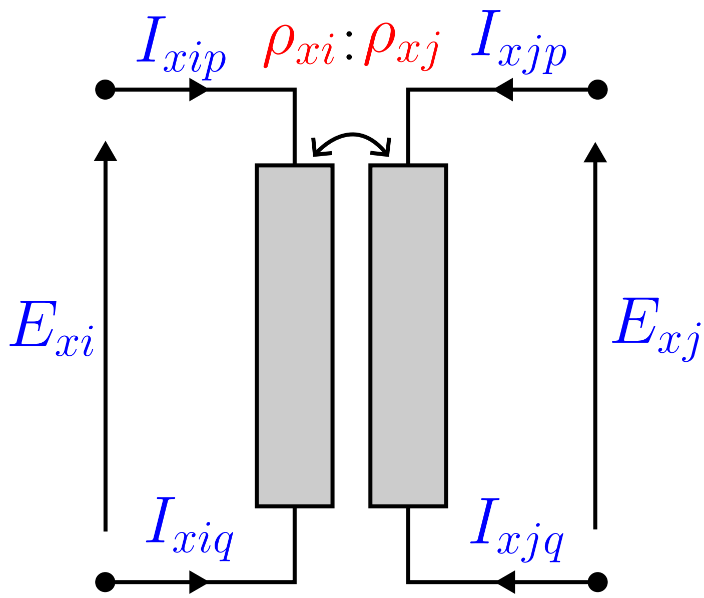

Figure 17.1: Idealised single-phase transformer: a pair of magnetically coupled windings.

how the windings connect.

- **Series leakage.** Each winding carries a series impedance between its EMF and its terminals (Ohm's law). Referred to the HV (from) side and combined, $Z_x = Z_x^{\text{fr}} + N_{\text{eff}}^2 Z_x^{\text{to}}$, the ideal voltage relation becomes $V_x^{\text{fr}} - N_{\text{eff}} V_x^{\text{to}} = Z_x I_{x,\text{fr}}$.
- **No-load (magnetising) shunt.** A single $Y_0 = G_0 + jB_0$ sits across **winding 2** (the to-side coil), referred to that coil's voltage, adding a current $Y_0 V_x^{\text{to}}$ to the to-side terminal — the core-loss/excitation branch (OpenDSS places it on winding 2, verified against its `Yprim`).
- **Neutral grounding.** When a wye winding's shared neutral terminal is earthed through an impedance, an internal branch $y_n = 1/(R_n + jX_n)$ draws $y_n U_{b,n}$ from that neutral terminal to earth.

The loss equivalent circuit

Read on the standard per-winding-pair equivalent circuit — shown here for the archetypal two-winding (single-phase) transformer, the reference picture every subtype below specialises:

- **Winding series impedance** — each coil carries a series leakage, $Z_x^{\text{fr}} = R_x^{\text{fr}} + jX_x^{\text{fr}}$ (from) and Z_x^{to} (to). This is the copper/leakage loss.
- **Short-circuit impedance** — a short-circuit test shorts one side and energises the other, so it measures the *series sum* of the two leakages referred to one side, $Z_{\text{sc}} = Z_x^{\text{fr}} + N_{\text{eff}}^2 Z_x^{\text{to}}$. It is **not** a separate element, and the split between the two windings is a *modelling choice*, since the test fixes only the sum (see the note below).
- **No-load loss and magnetisation** — the shunt $Y_0 = G_0 + jB_0$ on the winding-2 coil carries the **core (no-load) loss** G_0 and the **magnetising** susceptance B_0, as an open-circuit test measures.

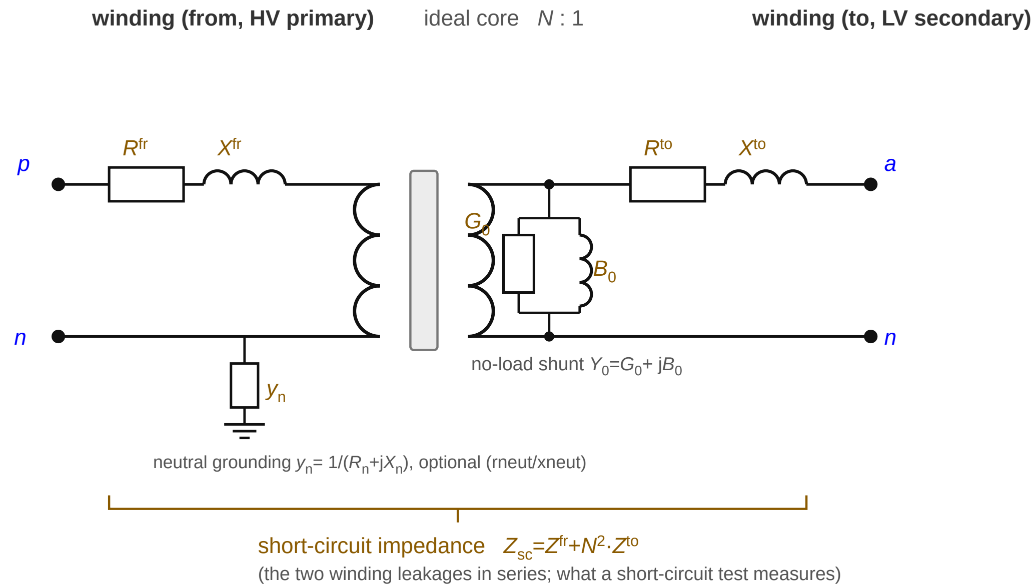

Figure 17.2: Two-winding (single-phase) loss equivalent circuit: a series winding impedance on each coil about the ideal core, their referred series sum forming the short-circuit impedance, and a no-load shunt (core loss + magnetisation) on the secondary coil.

The **optional neutral-grounding branch** $y_n = 1/(R_n + jX_n)$ (shown on the from-side neutral) is the third loss element: an internal branch from a wye winding's neutral terminal to earth. These map one-to-one onto the data fields: `r/x_series_from` → Z_x^{fr}, `r/x_series_to` → Z_x^{to}, `g_no_load` → G_0, `b_no_load` → B_0, `r/x_neutral_from/_to` → y_n. The subtype diagrams below omit the y_n branch to keep the connection clear; it attaches to whichever winding carries a groundable neutral (`single_phase`, `center_tap`, `wye_delta/delta_wye`).

The per-winding leakage split is under-determined by a short-circuit test

A standard short-circuit test yields only Z_{sc} — the *sum* of the two winding leakages. Splitting it into Z_x^{fr} and Z_x^{to} requires an extra convention (OpenDSS splits per its winding definitions; a common default is to put it all on one winding, i.e. the Γ-model with the other winding's leakage zero). The data model exposes both fields so the convention is explicit rather than assumed.

The subtypes below re-arrange exactly these three elements: **single-phase** is the picture above; **centre-tap** replaces the single secondary arm with two LV-leg arms; **wye-delta / delta-wye** wrap the pair in a Δ/Y connection with a $\sqrt{3}$ referral; and **n-winding** generalises the two leakage arms to a star.

Single-phase (wye-wye)

One winding pair per phase — the archetypal two-winding transformer, whose loss equivalent circuit is the canonical one in The loss equivalent circuit above. With the combined leakage $Z_x = Z_x^{\text{fr}} + N_{\text{eff}}^2 Z_x^{\text{to}}$:

$$V_{x,k}^{\text{fr}} - N_{\text{eff}} V_{x,k}^{\text{to}} = Z_x I_{x,\text{fr},k}, \qquad N_{\text{eff}} I_{x,\text{fr},k} + I_{x,\text{to},k} = 0.$$

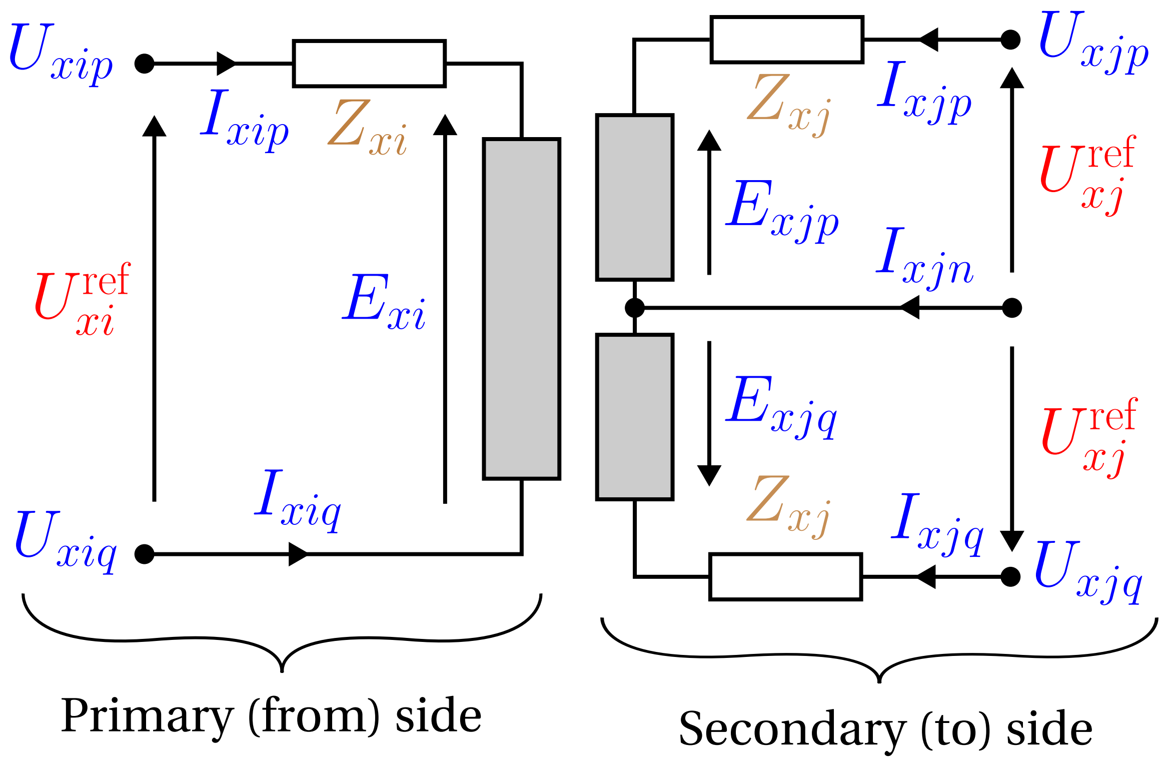

Figure 17.3: Centre-tap transformer: one HV winding, two anti-series LV legs sharing a centre-tap neutral.

The to-side terminal current is $I_{x,\mathsf{to},k} + Y_0 V^{\mathsf{to}}_{x,k}$ (series + magnetising); the neutral grounding branch, if present, adds its current at the shared neutral.

Center-tap (split-phase)

One HV winding drives **two anti-series LV legs** sharing a centre-tap neutral: two from terminals $[t^{\mathsf{ph}}, t^{\mathsf{n}}]$, three to terminals $[t_1, t^{\mathsf{n}}, t_2]$. The two half-windings are tightly coupled, so the model is a genuine three-winding (coupled-coil) unit, not two independent legs.

Writing $V^{\mathsf{hv}} = U_{i,t^{\mathsf{ph}}} - U_{i,t^{\mathsf{n}}}$, $v_1 = U_{j,t_1} - U_{j,t^{\mathsf{n}}}$, $v_2 = U_{j,t^{\mathsf{n}}} - U_{j,t_2}$ (winding 3 dotted at the centre tap), HV series current I_s and leg currents $I_{\ell 1}, I_{\ell 2}$:

$$
\begin{aligned}
V^{\mathsf{hv}} - N_{\mathsf{eff}}\, v_1 &= Z^{\mathsf{fr}}_x\, I_s - N_{\mathsf{eff}}\, Z^{\mathsf{to}}_x\, I_{\ell 1},\\
V^{\mathsf{hv}} - N_{\mathsf{eff}}\, v_2 &= Z^{\mathsf{fr}}_x\, I_s + N_{\mathsf{eff}}\, Z^{\mathsf{to}}_x\, I_{\ell 2},
\end{aligned}
$$

with the ampere-turn coupling and centre-tap KCL

$$
N_{\mathsf{eff}}\, I_s + I_{\ell 1} - I_{\ell 2} = 0, \qquad I_n + I_{\ell 1} + I_{\ell 2} = 0,
$$

where I_n is the centre-tap current (the leg imbalance). The HV series current returns through the HV neutral, $I_{x,\mathsf{fr},n} = -I_s$. The magnetising shunt Y_0 sits across winding 2 (LV leg 1, v_1). The $\mp N_{\mathsf{eff}} Z^{\mathsf{to}}_x$ sign difference between the legs is the reversed dotting of winding 3; using $+$ for both makes the legs identical and loses the load-imbalance physics.

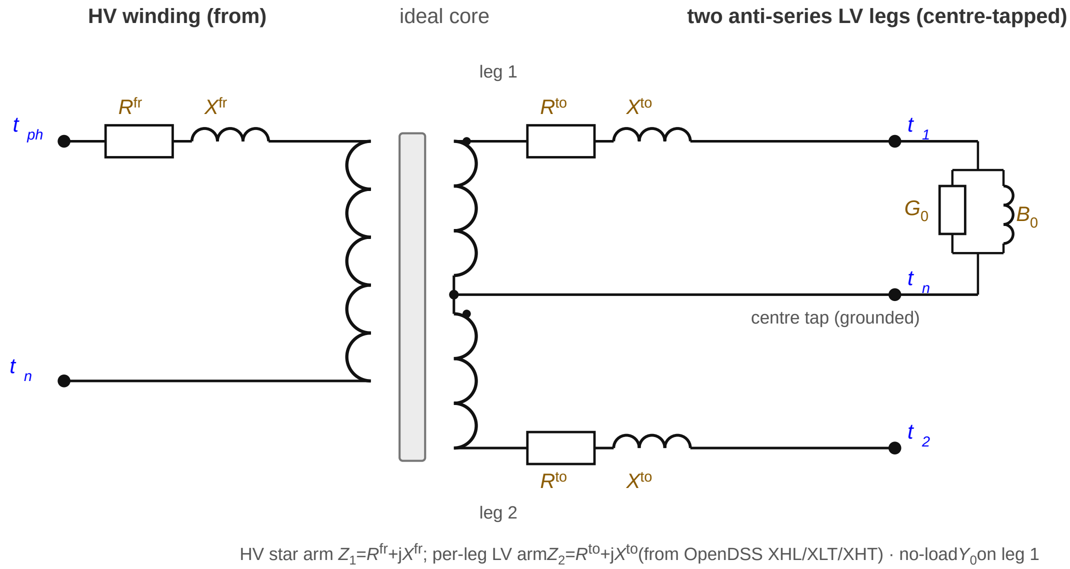

Figure 17.4: Centre-tap loss equivalent circuit: an HV winding series impedance and a series impedance on each of the two anti-series LV legs, with a no-load shunt (core loss + magnetisation) on leg 1.

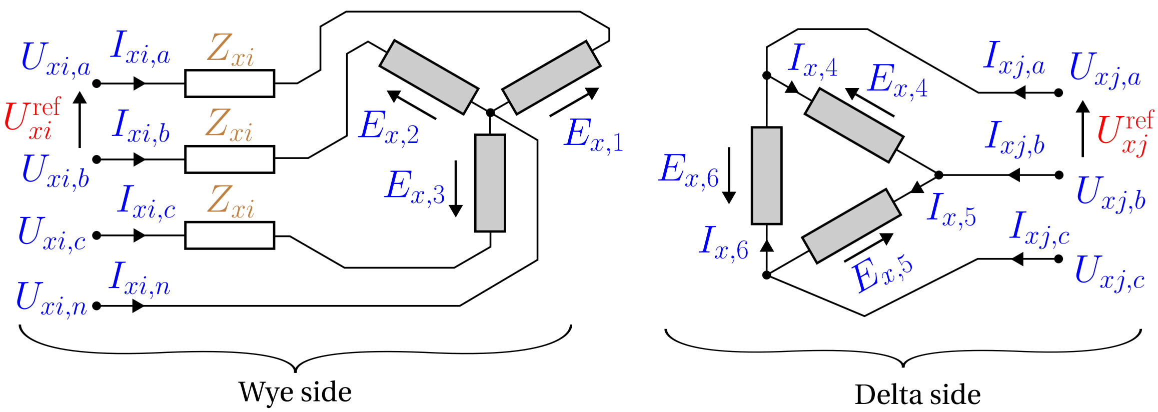

Figure 17.5: Three-phase wye–delta (or delta–wye) transformer connection and winding variables.

The same three loss elements (The loss equivalent circuit), but as a genuine three-winding unit its leakage is a **star** of arms — an HV arm and one per LV leg, not a single series pair — with the no-load shunt on leg 1:

Wye–delta and delta–wye

Three winding pairs. The delta connection introduces a $\sqrt{3}$ factor, so the effective per-winding ratio is

$$n^{\mathrm{eff}} = \begin{cases} \sqrt{3}/N_{\mathrm{eff}} & \text{wye\_delta (wye is from)}, \\ N_{\mathrm{eff}}\sqrt{3} & \text{delta\_wye (delta is from)}. \end{cases}$$

For phase k with cyclic partner k' (next for Yd, previous for Dy), the delta line-to-line voltage equals n^{eff} times

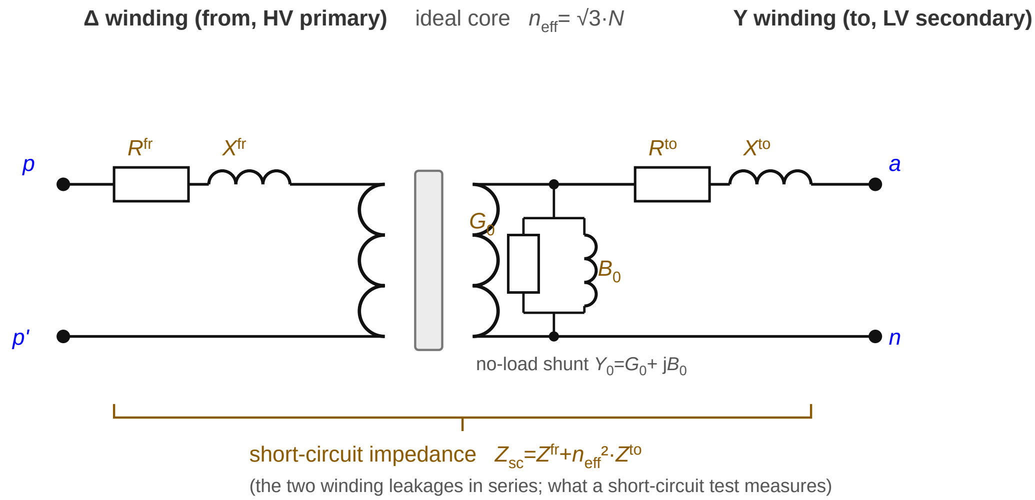

Figure 17.6: Delta–wye per-winding-pair loss equivalent circuit: a delta primary coil and wye secondary coil about the ideal core, each with its series winding impedance; the two leakages in series form the short-circuit impedance, and a no-load shunt (core loss plus magnetisation) hangs off the secondary coil.

the wye phase-to-neutral voltage, less the series drop on the wye phase current $I_{x,\mathrm{wye},k}$ through the effective impedance $Z_{\mathrm{eff}} = n^{\mathrm{eff}} Z_{\mathrm{wye}} + n_\phi Z_{\mathrm{del}}$:

$$U_{\mathrm{del},k} - U_{\mathrm{del},k'} = n^{\mathrm{eff}}\big(U_{\mathrm{wye},k} - U_{\mathrm{wye},n}\big) - Z_{\mathrm{eff}}\, I_{x,\mathrm{wye},k}.$$

The current transform is the transpose (power-conservative), $n^{\mathrm{eff}} I_{x,\mathrm{del},k} = -(I_{x,\mathrm{wye},k} - I_{x,\mathrm{wye},k'})$, and the wye star point satisfies $I_{x,\mathrm{wye},n} + \sum_k I_{x,\mathrm{wye},k} = 0$. The magnetising shunt sits across the winding-2 coils (a delta of branches when the delta is winding 2, phase-to-neutral when the wye is winding 2); the wye neutral may be grounded through y_n.

The same three loss elements (The loss equivalent circuit) here wrap a delta primary and a wye secondary, with the $\sqrt{3}$ folded into the effective ratio n^{eff}:

General n-winding

The `n_winding` object models an arbitrary number of windings (each wye or delta) on a shared core, from the OpenDSS-style short-circuit reactance matrix $\mathbf{Z}_B$ referred to **winding 1**. With per-winding turns ratio $N_k = U_k^{\mathrm{ref}}/U_1^{\mathrm{ref}}$ ($N_1 = 1$), referred coil currents $I_k^{\mathrm{r}} = N_k I_k$ and referred coil voltages $V_k^{\mathrm{r}} = U_k/N_k$, per phase/leg:

$$\sum_{k=1}^{n} N_k\, I_k = 0, \qquad V_1^{\mathrm{r}} - V_{i+1}^{\mathrm{r}} = \sum_{j=1}^{n-1} Z_{B,ij}\, I_{j+1}^{\mathrm{r}}, \quad i = 1, \ldots, n-1.$$

The coil voltage U_k is phase-to-neutral for a wye winding and line-to-line for a delta winding (whose `v_ref` is the line-to-line coil voltage, so the $\sqrt{3}$ lives in N_k). Each wye coil returns its phase currents through its neutral;

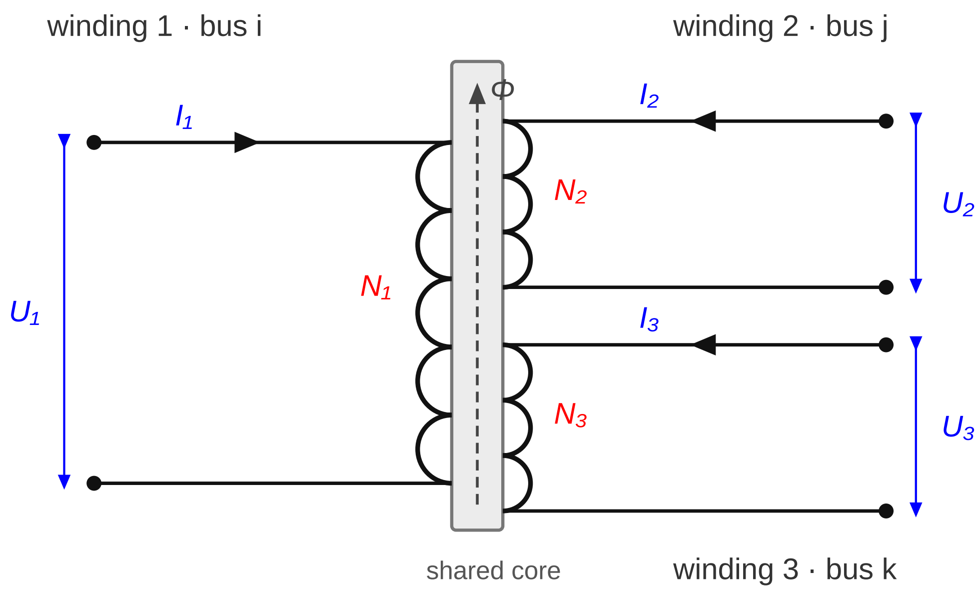

Figure 17.7: Multi-winding transformer: several windings on one shared magnetic core, each on its own bus, coupled by the common flux Φ; winding k carries current I_k at turns ratio N_k.

each delta coil injects between its two phase nodes. The magnetising shunt again sits across winding 2's coil. Tap optimisation is **not** supported for `n_winding` (the ratios are fixed).

This is the general form of the loss model: the leakage is a **star of per-winding arms** (referred to winding 1) meeting at a common core node, with the no-load shunt at that node. It generalises the two-winding picture — where the two arms in series *are* the single short-circuit impedance — to n windings, where each unordered pair (i, j) has its own short-circuit reactance $x_{\text{sc}}[i, j]$ (the field `x_sc` keyed `"i_j"`), and the star arms are recovered from that matrix ($x_{\text{sc}}[i, j] = X_i + X_j$ for a symmetric star):

So for more than two windings there is **no single "short-circuit reactance"** — only a matrix of pairwise short-circuits, which is exactly what the `x_sc` field records.

5. Inequality constraints

Cartesian variable bounds

Optional per-conductor current boxes on the winding-current components, from `i_max_from` / `i_max_to` (`i_max` per winding for `n_winding`) — implied by the current circles below.

Engineering bounds

Per-winding current-magnitude circles, per conductor k and side σ:

$$I_{x,\sigma,k}\,(I_{x,\sigma,k})^* \le (I^{\max}_{x,\sigma,k})^2.$$

Nameplate power. The winding-pair power transfer is bounded by the rating, $|E^{\text{fr}}_x\,(I_{x,\text{fr}})^*| \le S^{\max}_x / n_x$ with n_x the number of winding pairs (1 single-phase, 3 three-phase; centre-tap uses 1 on the from winding and 2 on the to legs).

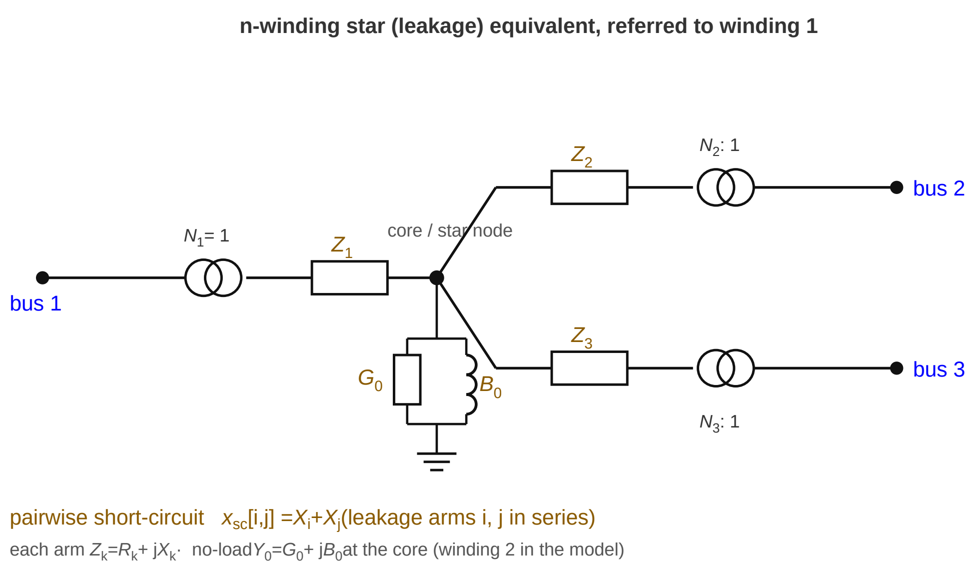

Figure 17.8: n-winding star (leakage) equivalent, referred to winding 1: each winding contributes a leakage arm to a common core node, the no-load shunt sits at the core, and each winding pair (i,j) has its own short-circuit reactance x_sc[i,j] = X_i + X_j.

Part IV

Objective & feasibility

Chapter 18

Objective and feasibility

The component pages define the network's variables and constraints — the *feasible set*. This page defines what is optimised over it: the **objective**, and the **feasibility relaxation** used to diagnose networks that have no feasible point. Unlike the component pages this is a *formulation* page, not a data object. Symbols are defined in Notation.

18.1 Objective

The default snapshot objective minimises total **active-power dispatch cost rate**, summed over every dispatchable element (generators, the voltage source, IBRs) and every phase:

$$\min \sum_e \sum_k c_{e,k} \; P_{e,k}/1000,$$

where $c_{e,k}$ (currency/kWh, from each element's per-phase `cost` array) is the energy price of phase k, and $P_{e,k}$ is that phase's injected active power in watts — the same bilinear expression the element defines ($P_{e,k} = \Delta v^r \, c^r + \Delta v^i \, c^i$). The factor $1/1000$ converts W to kW, so this snapshot objective is a cost **rate** in currency/h. For a multi-period monetary objective, multiply every snapshot rate by its duration in hours before summing.

Sign convention

$$P_{e,k}$$

is the power **injected into the network** by the element, uniformly across generators, IBRs, and the voltage source (each stamps $+I$ into KCL). Therefore:

- a **positive** cost minimises that element's injection;
- a **negative** cost maximises it.

The voltage source is *not* special: for the slack, positive injection means importing from the grid, so a positive source cost is the grid import price (and export, a negative injection, is credited at the same price). Maximising system exports is a positive slack cost with free DERs.

The cost is **linear** in the dispatch and is added exactly — there is no polynomial/quadratic term. A `cost` must be a per-phase vector; a scalar is rejected.

18.2 Feasibility relaxation

A constant-power OPF can be **infeasible**: the load/generation specification may be irreconcilable with the network physics. To diagnose such cases, a feasibility-relaxed variant adds an **elastic slack current** and minimises it, instead of cost.

Elastic slack

At every ungrounded, non-source terminal a free slack current $s_{i,p} = s^r_{i,p} + j\, s^i_{i,p}$ is added directly into that terminal's KCL:

$$\kappa^{\Re}_{i,p} + s^r_{i,p} = 0, \qquad \kappa^{\Im}_{i,p} + s^i_{i,p} = 0.$$

Where a slack pair is present, it can absorb any residual in that terminal's KCL. This does **not** guarantee that the complete relaxed NLP is feasible or that a local solver will converge: fixed source voltages, hard bounds, device equalities, or terminals without slacks can still conflict.

Objective

The cost objective is replaced by the squared magnitude (ℓ_2^2) of all slack injections:

$$\min \sum_{i,p} \big((s^r_{i,p})^2 + (s^i_{i,p})^2\big).$$

Interpretation

The relaxation retains the standard OPF's hard constraints — voltage bounds, bus/line angle limits, and every device current limit — while enlarging the feasible set only through the nodal-current slacks. Consequently, for a successfully converged solve:

- An independently residual-checked **zero-slack** point demonstrates numerical feasibility under the represented loading and constraints.
- **Non-zero** slack at a locally optimal relaxed point identifies where that solve paid to violate KCL and by how much current. It is diagnostic evidence, not a proof that the original nonconvex problem has no zero-slack solution.
- Voltages still respect their hard bounds. When those hard constraints remain mutually consistent, an unreachable operating bound commonly surfaces as residual current. Contradictory hard constraints can instead leave even the relaxed problem infeasible (for example, a fixed source voltage contradicting a bound on the same terminal).

Part V

Reference

Chapter 19

Modelling notes & FAQ

19.1 Modelling notes and FAQ

Practical questions that recur when building cases. Symbols are defined in Notation.

How do I model a constant-impedance or constant-current load?

Use the load `model` field: `constant_impedance` ($P \propto |V|^2$), `constant_current` ($P \propto |V|$), or a full `zip` mix, with `v_nom` giving the reference voltage. (Older guidance suggested emulating a constant-impedance load with a shunt; that is no longer necessary now that voltage-dependent load models are supported.)

How do I define a triplex (split-phase service) load?

As one or more **single-phase** loads across the triplex terminals. For a 1 kW load across legs 1–2 of a triplex bus with `terminal_names ["1","n","2"]`, define a `SINGLE_PHASE` load with `terminal_map ["1","2"]` and `p_nom [1000.0]`. A two-terminal `SINGLE_PHASE` map is modelled across exactly those two terminals (here line-to-line, 240 V), not phase-to-ground — see Loads.

I converted a wye load to delta with the standard transform and got a different answer. Why?

Because a distribution **wye load is four-wire** — three phase branches to a *neutral return* conductor — whereas the textbook delta–wye (Y–Δ) transform assumes a *three-wire* wye with no return (a graph-theoretic star). The transform is **not applicable** to a wye-with-neutral load. Model the wye and delta connections directly via the load `configuration` field; do not pre-transform.

How do I model a three-wire wye load (no neutral return)?

Add a **floating midpoint** terminal at the bus and connect three single-phase loads (or a wye-with-return whose neutral ties to that floating node) across the phases to it. The floating node carries no external connection, so it enforces the zero-return-current condition of a true three-wire star. There is no dedicated "wye-without-neutral" load subtype.

How do I get center-tap transformer impedances from OpenDSS or Gridlab-D?

OpenDSS specifies the three inter-winding short-circuit reactances $\texttt{Xhl}, \texttt{Xlt}, \texttt{Xht}$ (% p.u.). These map to the per-winding reactances by the star (T) transform $[\texttt{Xh}; \texttt{Xl}; \texttt{Xt}] = \frac{1}{2}\mathbf{S}\,[\texttt{Xhl}; \texttt{Xlt}; \texttt{Xht}]$ with $\mathbf{S} = \begin{bmatrix} 1 & -1 & 1 \\ 1 & 1 & -1 \\ -1 & 1 & 1 \end{bmatrix}$, then to ohms via the winding voltage/power base. The transformer page gives the full partitioning (and warns against the common `Xhl`$/2$ shortcut, which is wrong for center-tap under unbalance).

Gridlab-D gives per-unit primary `impedance` and secondary `impedance1` with a `power_rating` for the *whole* transformer. The per-winding SI impedances are

$$Z_i = \texttt{impedance} \cdot \frac{3}{1000} \cdot \frac{\texttt{primary\_voltage}^2}{\texttt{power\_rating}}, \qquad Z_j = \texttt{impedance1} \cdot \frac{3}{1000} \cdot \frac{\texttt{secondary\_voltage}^2}{\texttt{power\_rating}},$$

where the factor 3 accounts for the total rating being three times the primary-winding power, and 1000 converts the kVA rating to VA.

Import tools usually apply these conversions

Tools that import from OpenDSS or Gridlab-D typically apply these conversions automatically when reading source models. The formulas are given here for authors constructing data by hand or auditing an import.

My line impedances are sequence components only — can I convert them to wire coordinates?

Yes. Sequence-component impedances convert to phase (wire) coordinates through the Fortescue transform $\mathbf{F}$ (defined in Notation). Writing sequence voltages and currents as $\mathbf{U}_i^{\mathsf{sym}} = \mathbf{F}\,\mathbf{U}_i[\mathcal{P}]$ and $\mathbf{I}_{\ell ij}^{\mathsf{sym,s}} = \mathbf{F}\,\mathbf{I}_{\ell ij}^{\mathsf{s}}[\mathcal{P}]$, and premultiplying the phase Ohm's law $\mathbf{U}_j[\mathcal{P}] = \mathbf{U}_i[\mathcal{P}] - \mathbf{Z}_\ell^{\mathsf{s}}\,\mathbf{I}_{\ell ij}^{\mathsf{s}}[\mathcal{P}]$ by $\mathbf{F}$ (inserting $\mathbf{I} = \mathbf{F}^{-1}\mathbf{F}$) gives the sequence-domain impedance

$$\mathbf{Z}_\ell^{\mathsf{sym,s}} = \mathbf{F}\,\mathbf{Z}_\ell^{\mathsf{s}}\,\mathbf{F}^{-1} = \begin{bmatrix} Z_{\ell,00} & Z_{\ell,01} & Z_{\ell,02} \\ Z_{\ell,10} & Z_{\ell,11} & Z_{\ell,12} \\ Z_{\ell,20} & Z_{\ell,21} & Z_{\ell,22} \end{bmatrix}.$$

Lines are commonly assumed transposed, so the off-diagonal terms are neglected and the positive and negative sequences take equal value, leaving $\mathbf{Z}_\ell^{\mathsf{sym,s}} \approx \mathrm{diag}(Z_{\ell,00}, Z_{\ell,11}, Z_{\ell,11})$. Inverting the transform then recovers the phase impedance matrix from just the zero- and positive-sequence values $Z_{\ell,00}, Z_{\ell,11}$:

$$\mathbf{Z}_\ell^{\mathsf{s}} = \frac{1}{3}\begin{bmatrix} 2Z_{\ell,11} + Z_{\ell,00} & Z_{\ell,00} - Z_{\ell,11} & Z_{\ell,00} - Z_{\ell,11} \\ Z_{\ell,00} - Z_{\ell,11} & 2Z_{\ell,11} + Z_{\ell,00} & Z_{\ell,00} - Z_{\ell,11} \\ Z_{\ell,00} - Z_{\ell,11} & Z_{\ell,00} - Z_{\ell,11} & 2Z_{\ell,11} + Z_{\ell,00} \end{bmatrix}.$$

Here $\mathbf{Z}_\ell^{\mathsf{s}}$ is the (three-wire, or Kron-reduced four-wire) $3{\times}3$ series-impedance matrix used as the line's `R_series`/`X_series`.

Chapter 20

Nomenclature

20.1 Nomenclature

A consolidated list of the recurring symbols. The typography and colour convention (variable vs parameter, real vs complex) and the transform constants are defined on the Notation page; the string-valued parameters and their permitted values are listed under Data input formatting; and the element sets, topology, connectivity, and terminal mappings are collected in Notation → Sets and indices.

Matrix dimensions $n \times n$ have $1 \leq n \leq 4$ for a four-wire, three-phase network.

Complex-valued variables

Symbol	Unit	Length	Represents
$\mathbf{U}_i$	V	4	Bus i phase-to-ground voltage vector
$\mathbf{U}_i^{\Delta}$	V	3	Bus i phase-to-phase voltage vector
$\mathbf{U}_i^{\text{pn}}$	V	3	Bus i phase-to-neutral voltage vector
$\mathbf{U}_i^{012}$	V	3	Bus i symmetrical-component voltage vector
$\mathbf{I}_{\ell ij}$	A	4	Line ℓ total current, direction $i \to j$
$\mathbf{I}_{\ell ij}^{\text{s}}$	A	4	Line ℓ series current, direction $i \to j$
$\mathbf{I}_{\ell ij}^{\text{sh}}$	A	4	Line ℓ shunt current, direction $i \to j$
$\mathbf{S}_{\ell ij}$	W	4	Line ℓ total power, direction $i \to j$
$\mathbf{S}_{\ell ij}^{\text{s}}$	W	4	Line ℓ series power, direction $i \to j$
$\mathbf{S}_d$	W	4	Load d power at its bus
$\mathbf{S}_d^{\text{pn}}$	W	3	Load d power in wye (phase-to-neutral) frame
$\mathbf{S}_d^{\Delta}$	W	3	Load d power in delta frame
$\mathbf{I}_d$	A	2, 3 or 4	Load d current at its bus
$\mathbf{I}_d^{\Delta}$	A	3	Load d internal delta current
$\mathbf{S}_g$	W	4	Generator g power at its bus
$\mathbf{S}_g^{\text{pn}}$	W	3	Generator g power in wye frame
$\mathbf{S}_g^{\Delta}$	W	3	Generator g power in delta frame
$\mathbf{I}_g$	A	4	Generator g current at its bus

Complex-valued parameters

Real-valued parameters

Symbol	Unit	Size	Represents
$\mathbf{Y}^{\text{sh}}_{\ell ij}$	S	$n \times n$	Line ℓ shunt admittance at sending end i
$\mathbf{Y}^{\text{sh}}_{\ell ji}$	S	$n \times n$	Line ℓ shunt admittance at receiving end j
$\mathbf{Z}^{\text{s}}_{\ell}$	Ω	$n \times n$	Line ℓ series impedance matrix
$\mathbf{Y}_{h}$	S	$n \times n$	Bus shunt admittance (neutral grounding, capacitor banks, ...)
$\mathbf{S}^{\text{ref,pn}}_{d}$	W+$j$var	3×1	Load d complex power set-point (wye-connected)
$\mathbf{S}^{\text{ref},\Delta}_{d}$	W+$j$var	3×1	Load d complex power set-point (delta-connected)
$\mathbf{S}^{\text{ref,pn}}_{g}$	W+$j$var	3×1	Generator g complex power set-point
$\bar{\mathbf{Z}}^{\text{s}}_{c}$	Ω/m	$n \times n$	Linecode per-length series impedance
$\bar{\mathbf{Y}}^{\text{sh}}_{c}$	S/m	$n \times n$	Linecode per-length shunt admittance

Symbol	Unit	Size	Represents
$\mathbf{U}^{\min}_{i}$, $\mathbf{U}^{\max}_{i}$	V	4×1	Bus i phase-to-ground magnitude bounds
$\mathbf{U}^{\text{pn},\min}_{i}$, $\mathbf{U}^{\text{pn},\max}_{i}$	V	3×1	Bus i phase-to-neutral magnitude bounds
$\mathbf{U}^{012,\min}_{i}$, $\mathbf{U}^{012,\max}_{i}$	V	3×1	Bus i symmetrical-component magnitude bounds
$\mathbf{I}^{\max}_{\ell ij}$	A	4×1	Line ℓ current-magnitude upper bound
$\mathbf{S}^{\max}_{\ell ij}$	VA	4×1	Line ℓ apparent-power upper bound
$\mathbf{P}^{\min}_{g}$, $\mathbf{P}^{\max}_{g}$	W	3×1	Generator g active-power bounds
$\mathbf{Q}^{\min}_{g}$, $\mathbf{Q}^{\max}_{g}$	var	3×1	Generator g reactive-power bounds
ℓ_{l}	m	scalar	Line length

Chapter 21

References & further reading

21.1 References and further reading

This specification stands on a large body of work in distribution-system modelling and optimal power flow. The lists below point to the textbooks that develop the underlying physics and the papers this model builds on directly. This is a *reference* page — it introduces no model content.

Textbooks

Distribution-system modelling (component models, Carson/Kron impedance, unbalanced analysis):

- **W. H. Kersting**, *Distribution System Modeling and Analysis*, CRC Press (4th ed., 2017). The standard reference for line/cable impedance (Carson's equations, Kron reduction), transformer connections, and unbalanced power flow — the physics behind the line and transformer pages.
- **T. A. Short**, *Electric Power Distribution Handbook*, CRC Press (2nd ed., 2014). A practical, equipment-oriented companion covering feeders, grounding, and protection.
- **R. C. Dugan, M. F. McGranaghan, S. Santoso, H. W. Beaty**, *Electrical Power Systems Quality*, McGraw-Hill (3rd ed., 2012). Background on unbalance, harmonics, and the power-quality phenomena that motivate conductor-level modelling. See also OpenDSS (Dugan/EPRI), the reference implementation this spec is validated against.

Optimal power flow and its mathematics:

- **S. H. Low**, *Power System Analysis: Analytical Tools and Structural Properties* (forthcoming graduate textbook). A rigorous, notation-first treatment of network models and OPF; freely available on registration at netlab.caltech.edu/book_reg. Its complex stacked-vector style is close to the notation used here.
- **S. H. Low**, "Convex Relaxation of Optimal Power Flow, Parts I & II," *IEEE Trans. Control of Network Systems*, 2014. Foundational for the relaxations that lift the current–voltage formulation used here to power/lifted-voltage spaces.

Accessible introductions and lecture notes

Approachable complements to the textbooks above for readers getting into distribution networks — and especially for building intuition about **transformer loss models** — where Steven Low's book covers the mathematics underneath:

- **Z. Wang**, distribution-systems course notes, Iowa State University: Introduction to Distribution Systems (EE455), Distribution System Transformers (EE555 — connections and loss models), and Real Distribution System Modeling and Analysis (EE653). Clear, worked treatments of feeder components and how transformer losses map onto the equivalent circuit.

- **S. Claeys, G. Deconinck, F. Geth**, "Decomposition of n-winding transformers for unbalanced optimal power flow," *IET Generation, Transmission & Distribution* **14**(24):5961–5969, 2020, doi:10.1049/iet-gtd.2020.0776 — a useful reference for conceptualising the transformer loss model.

Foundational papers for this model

The four-wire current–voltage (IVR-EN) formulation, its benchmarking, and the device models:

- **S. Claeys et al.**, "Optimal power flow in four-wire distribution networks: Formulation and benchmarking," *Electric Power Systems Research*, 2022. The four-wire OPF formulation this specification extends.

- **D. M. Fobes, S. Claeys, F. Geth, C. Coffrin**, "PowerModelsDistribution.jl: An open-source framework for exploring distribution power flow formulations," *Electric Power Systems Research*, 2020. The `IVRENPowerModel` lineage of the OPF engine.

- **F. Geth, H. Ergun**, "Real-Value Power-Voltage formulations of, and bounds for, three-wire unbalanced optimal power flow," 2023. The real-valued bound machinery behind the engineering bounds.

- **F. Geth et al.**, "Considerations and design goals for unbalanced optimal power flow benchmarks," *Electric Power Systems Research*, 2024. The benchmarking philosophy behind this Task Force effort.

- **R. C. Dugan**, "A perspective on transformer modeling for distribution system analysis," *IEEE PES General Meeting*, 2003. Background for the transformer winding models and grounding conventions.

Benchmark libraries and test systems

- **S. Babaeinejadsarookolaee et al.**, "The Power Grid Library for benchmarking AC optimal power flow algorithms" (PGLib-OPF), 2019. The transmission-side, positive- sequence analogue of what this Task Force provides for unbalanced distribution.

- **K. P. Schneider et al.**, "Analytic Considerations and Design Basis for the IEEE Distribution Test Feeders," *IEEE Trans. Power Systems*, 2018. The IEEE Distribution Test Feeder Working Group's library (including the 8500-node feeder).

Tools

- **OpenDSS** (EPRI) — a widely used distribution power-flow engine and a common source and cross-validation reference for distribution models.

- **PowerModelsDistribution.jl** — the Julia framework whose `IVRENPowerModel` inspired this formulation.

Chapter 22

Glossary

22.1 Glossary

A quick reference for the terms and abbreviations used throughout the specification. Symbols and mathematical notation are defined on the Notation page.

> **Work in progress**
>
> This glossary is a stub. Entries are added as terms are introduced across the specification — contributions welcome (see Contributing).

Terms

Four-wire model : A network representation in which each of the three phases **and** the neutral conductor are modelled explicitly, with earth as a separate return path.

Nodal admittance matrix (Ybus) : The matrix $\mathbf{Y}$ relating injected currents to node voltages, $\mathbf{I} = \mathbf{Y}\,\mathbf{V}$.

OPF — Optimal Power Flow : An optimisation problem that determines the operating point of a network minimising an objective (e.g. cost, losses) subject to physical and operational constraints. See Objective & feasibility.

ZIP load : A voltage-dependent load model combining constant-impedance (Z), constant-current (I), and constant-power (P) components. See Loads.